Modular Robots - Locomotion and Obstacle Avoidance



Avinash Ranganath

1. A brief introduction to the field of Reconfigurable Modular Robotics.

2. **Juan Gonzalez**´s presentation on his work in the field of Modular Robotics.

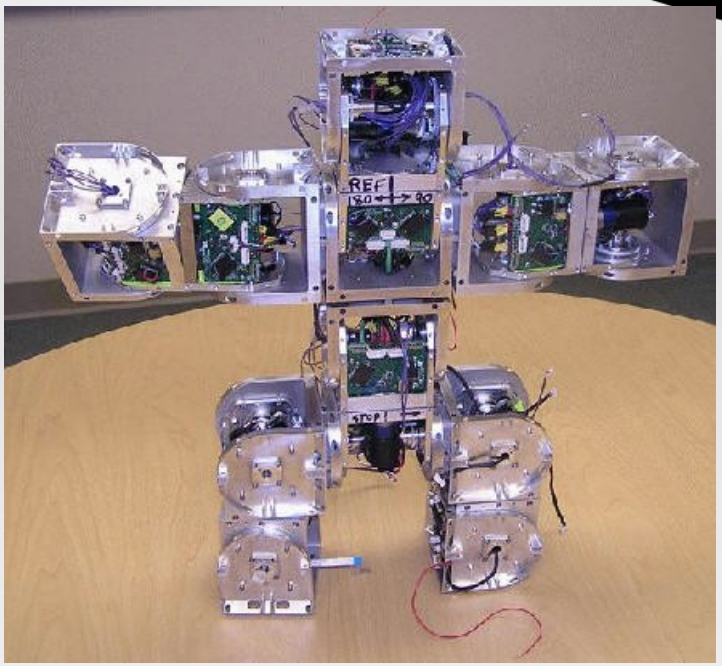
3. Presenting my master thesis work in Modular Robotics.

Supervisor: Marc Szymanski from IPR, KIT, Karlsruhe, Germany.

Supervisor: Barbara Webb from Edinburgh University, Edinburgh, Scotland.

Introduction – Reconfigurable Modular Robots

- What are Reconfigurable Modular Robots?
  - Made up of several independent modules.
  - Which does not have a fixed morphology.





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Introduction – Common Features of Reconfigurable Modular Robots



- Each module is independent with its own on-board processor, sensor, actuator and power supply.
- Each module can be connected to two or more modules.
- Ability of modules to connect to or disconnect from other modules autonomously.
- Ability to communicate with other modules.

Introduction – Types of Reconfigurable Modular Robots

- Chain Type



- Lattic Type



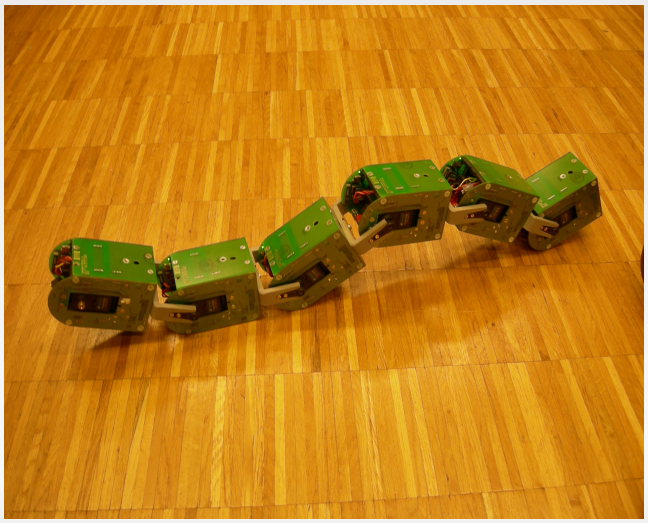
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Motivation – Reconfigurable Modular Robotics

- Exploration in unknown environment.
  - Outer space.
  - Collapsed building or disaster area.







Control Mechanism – Reconfigurable Modular Robotics

- Centralized Controller
  - Easy to implement.
  - Susceptible to the failure if the central controlling module fails.
  - Non scalable.
  - Under utilization of distributed computing capabilities.
- Distributed Controller
  - Complex and difficult to implement.
  - Robust
  - Scalable
  - Distributed computing capability without overloading any individual module.

Juan´s Presentation



Passing the baton to Juan...

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Locomoción de Robots ápodos modulares



Juan González Gómez

Dpto. Ingeniería de Sistemas y Automática
Robotics Lab

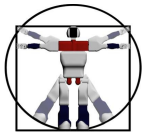
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4. Plataforma
5. Conclusiones y trabajo futuro

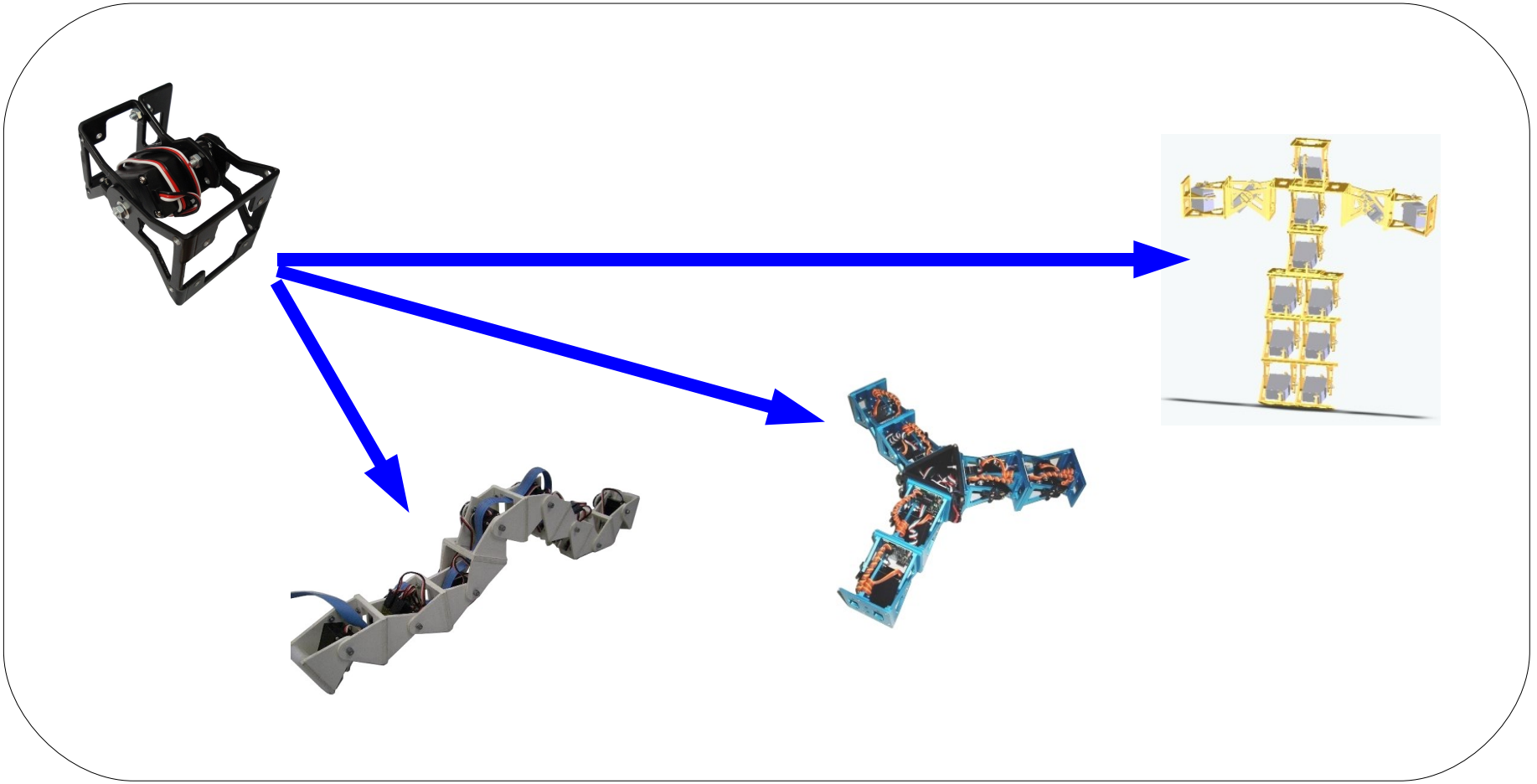




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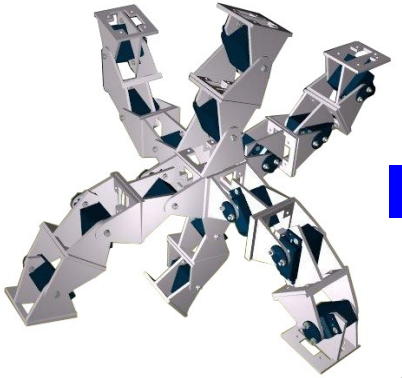
Robots modulares



Grupos de estudio

Robots modulares



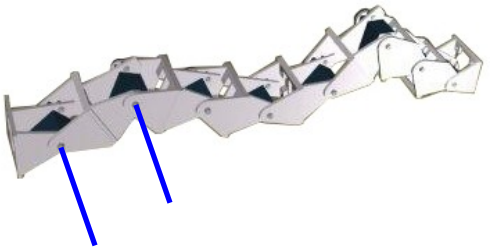


Topología 1D

Robots ápodos



Pitch-pitch



Locomoción 1D

Pitch-yaw



Locomoción 2D

Locomoción de robots ápodos

Controlador

¿Cómo coordinar las articulaciones para lograr la locomoción?

GAITs

¿Qué modos de caminar se consiguen?



Configuraciones mínimas

¿Cuántos módulos como mínimo se necesitan para lograr la locomoción?

CONTROL

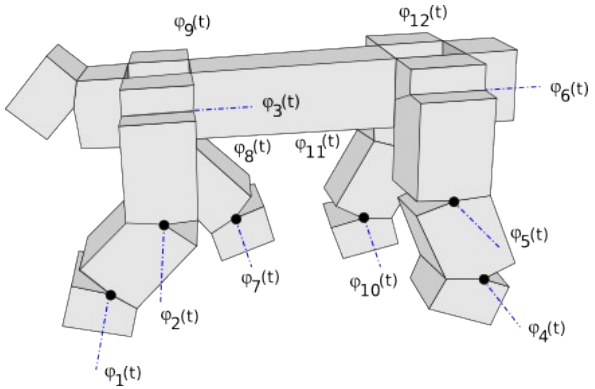
¿Cual es el espacio de control de menor dimensión ?

Controlador

Modelos para realizar la coordinación e implementar el controlador:

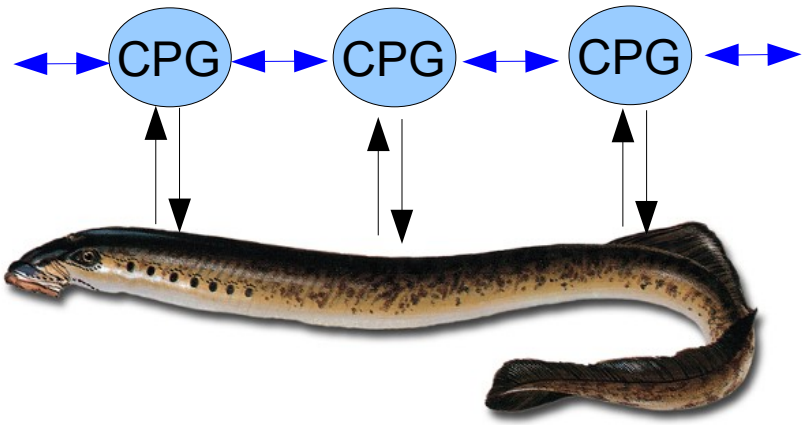
Clásicos

- Modelos matemáticos
- Cinemática inversa
- Dependen de la morfología del robot



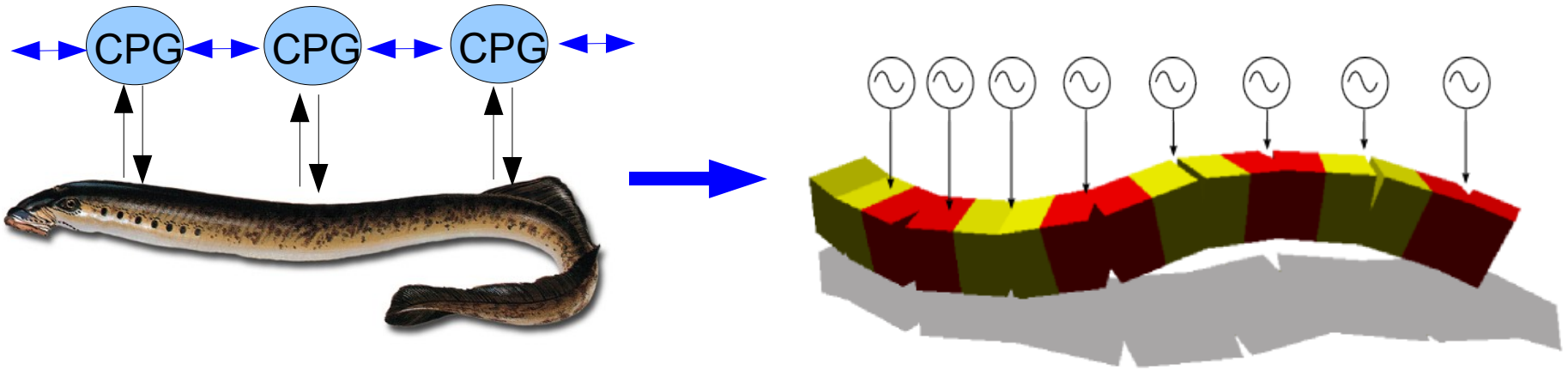
Bio-inspirados

- Imitar la naturaleza
- Generadores Centrales de patrones: CPG

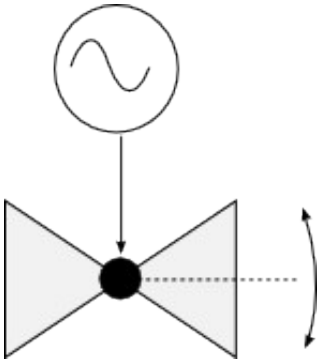


Osciladores sinusoidales

- Reemplazar los CPGs por **OSCILADORES SINUSOIDALES**



- Osciladores sinusoidales:



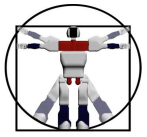
$$\varphi_i(t) = A_i \sin\left(\frac{2\pi}{T} t + \psi_i\right)$$

¿Es viable?

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2. **Locomoción en 1D**
3. Locomoción en 2D
4. Plataforma
5. Conclusiones y trabajo futuro





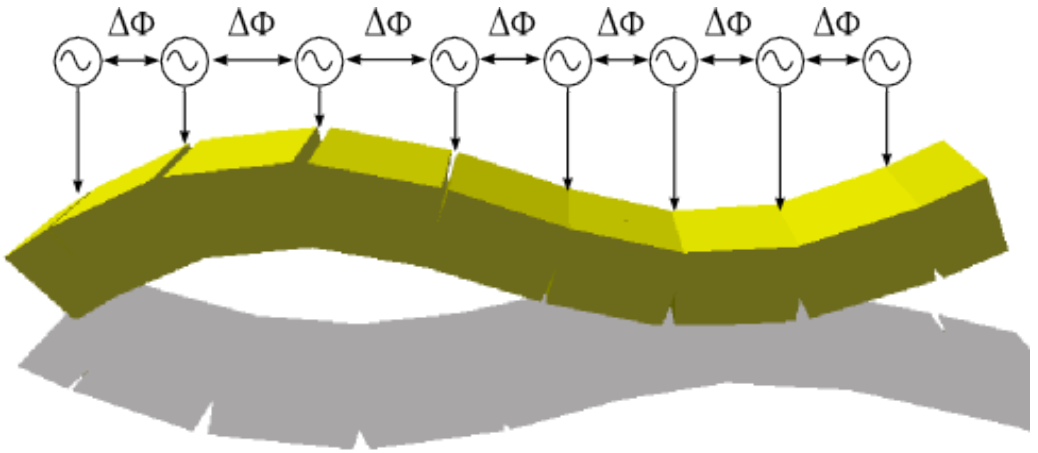
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Locomoción en 1D: Resultados

Vídeos 1-3

Modelo de control



- El modelo es viable
- Movimiento: Propagación de ondas. Adelante-Atrás
- Configuración mínima: 2 módulos
- Espacio de control mínimo: $A, \Delta\Phi, T$

Paso del robot

Nº ondulaciones

Velocidad

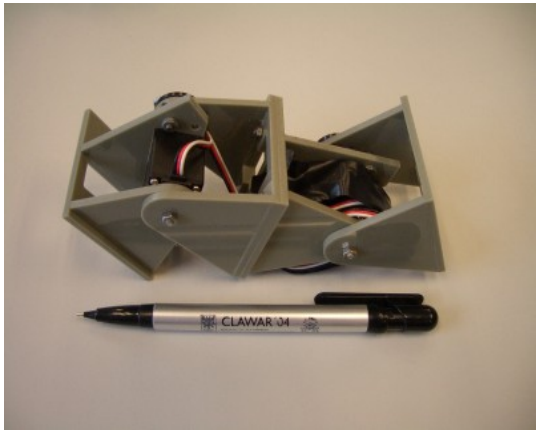
Minicube-I (II)

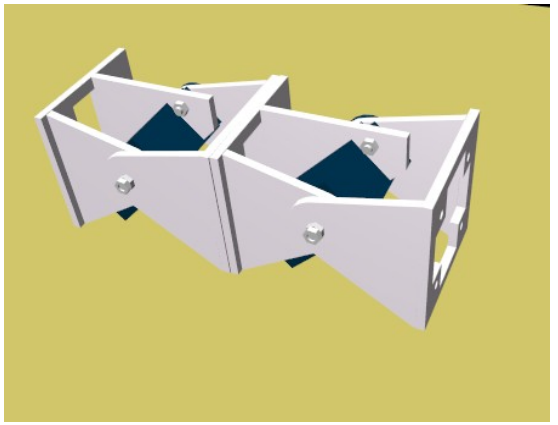
Demo

- **Morfología**

  2 modules con conexión cabeceo-cabeceo

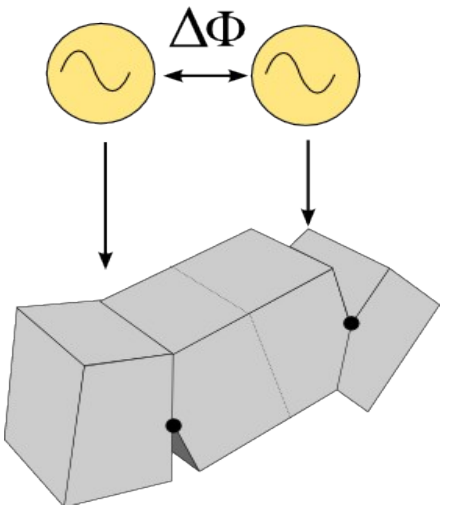
  Locomoción en 1D





- **Controlador:**
  - Dos generadores iguales
  - Parámetros $A, \Delta\Phi, T$
  - Más información:

    http://bit.ly/9SNFXb



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1. Introducción

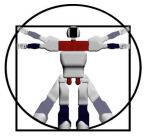
2. Locomoción en 1D

3. **Locomoción en 2D**

4. Plataforma

5. Conclusiones y trabajo futuro





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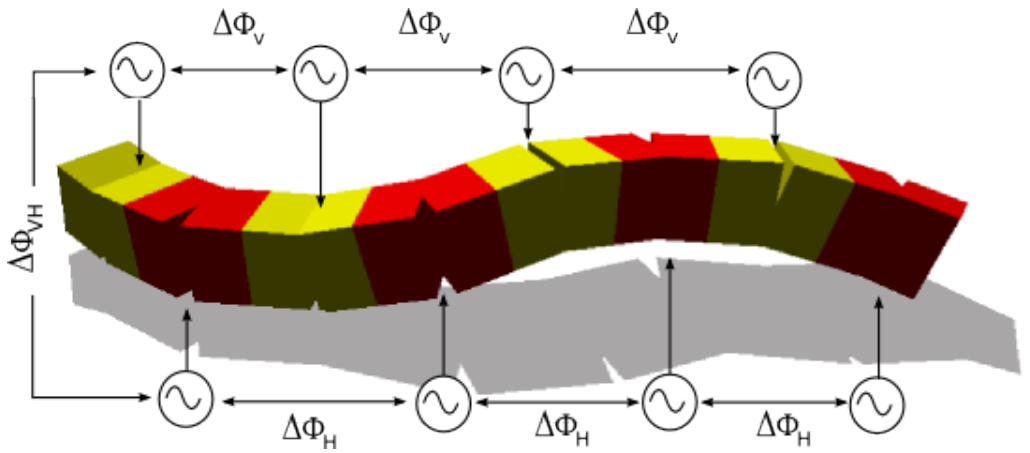
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Locomoción en 2D: Resultados

Vídeo 4

Modelo de control



- El modelo es viable
- 5 movimientos: línea recta, arco, lateral, rotar y rodar
- Configuración mínima: 3 módulos
- Espacio de control mínimo: $A_h, A_v, \Delta\Phi_h, \Delta\Phi_v, \Delta\Phi_{vh}, T$

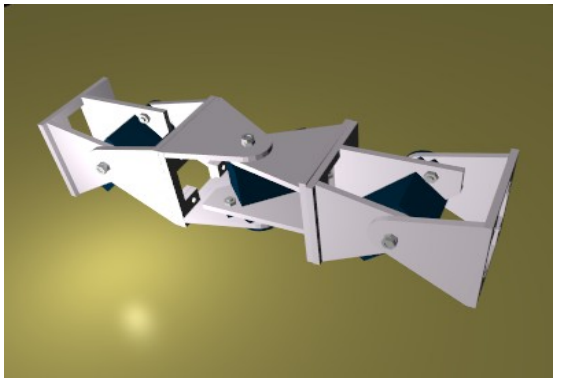
Minicube-II

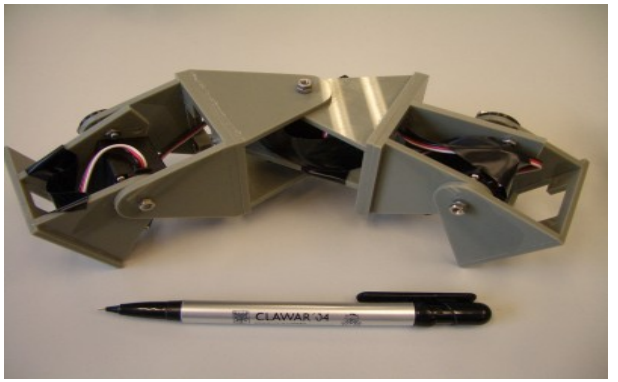
Demostración

- **Morfología:**

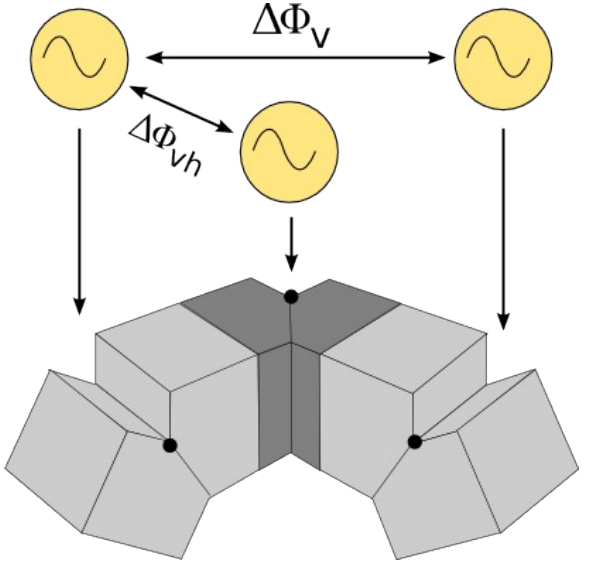
  Tres módulos con conexión cabeceo-viraje

  Locomoción en 2D





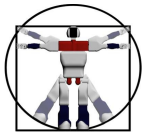
- **Control:**
  - Tres generadores sinusoidales
  - Parámetros: $A_v, A_h, \Delta\Phi_v, \Delta\Phi_{vh}, T$



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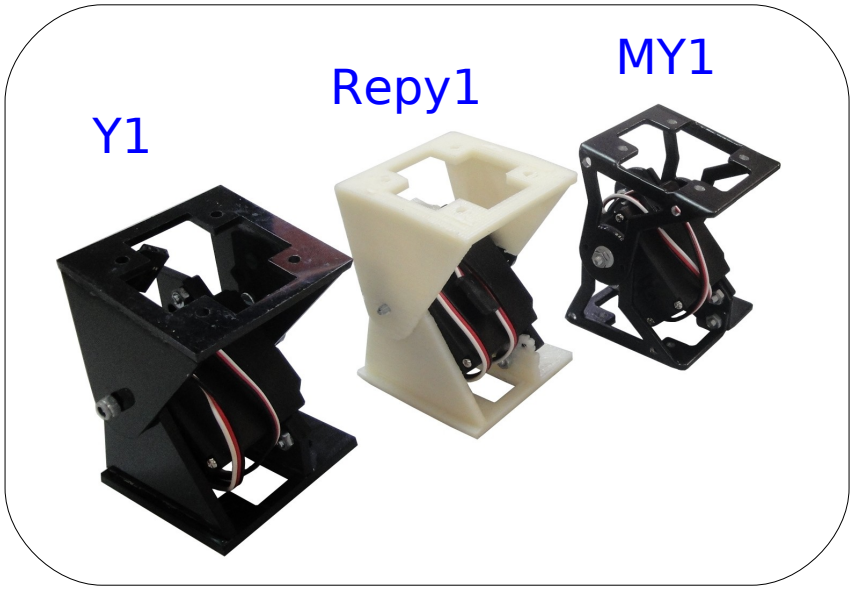


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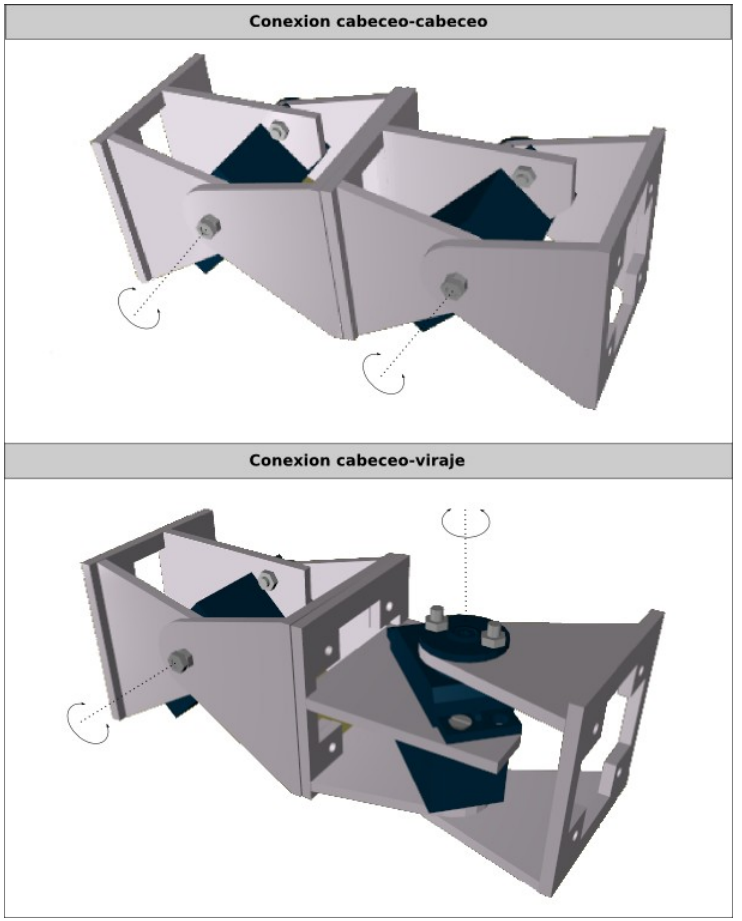
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Mecánica: Familia de Módulos Y1

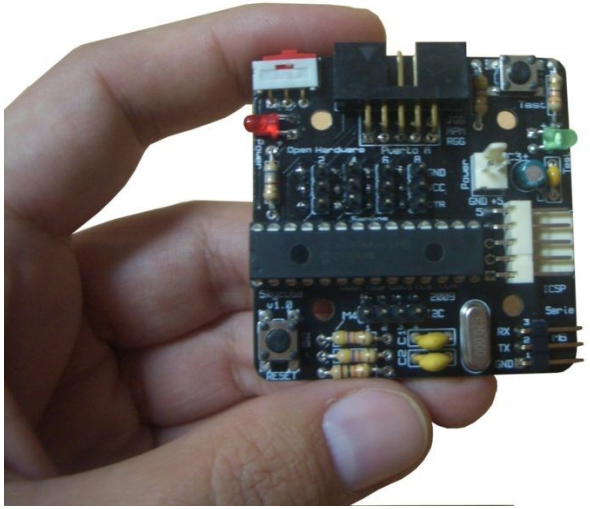
- Un grado de libertad
- Fáciles de construir
- Servo: Futaba 3003
- Tamaño: 52x52x72mm
- **Libres**

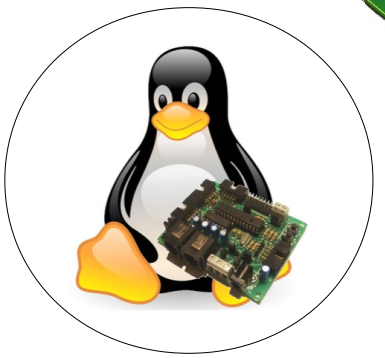


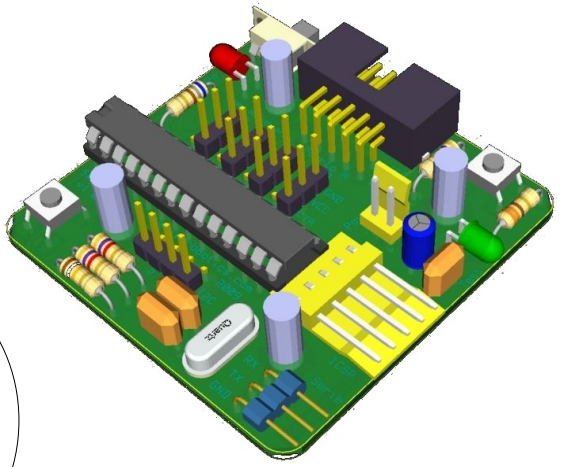
Tipos de conexión:



Electrónica: Tarjeta Skycube

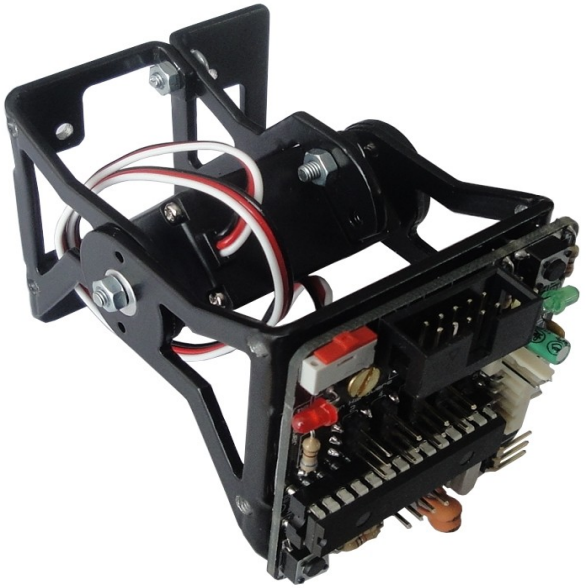






- **Hardware libre**
- Diseñada con KICAD
- Robots modulares autónomos
- PIC16F876A
- Se integra en los módulos MY1
- Más información:

http://bit.ly/FhPLI



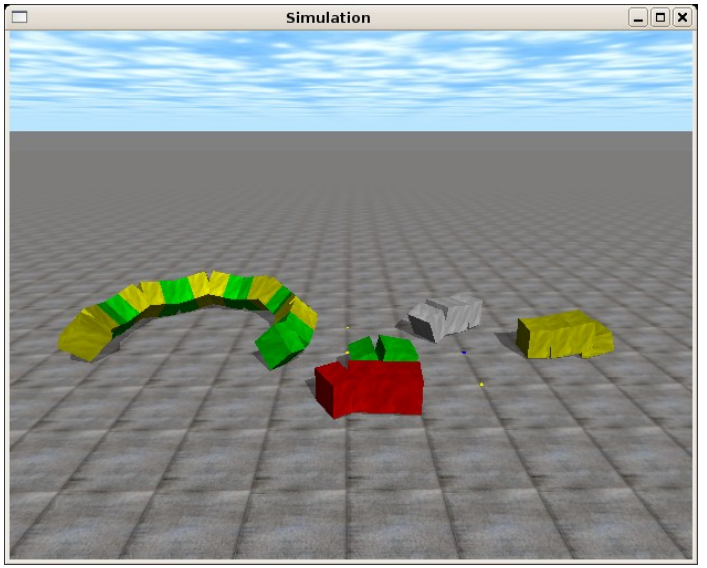
Simulación (I)

Demo

¿Cómo hemos encontrado las soluciones?

- Búsquedas en los espacios de control
- Utilización de algoritmos genéticos (PGApack)
- Función de evaluación: Paso del robot
- Motor físico: Open Dynamics Engine (ODE)
- Descarte de soluciones
- **Comprobación en robots reales**

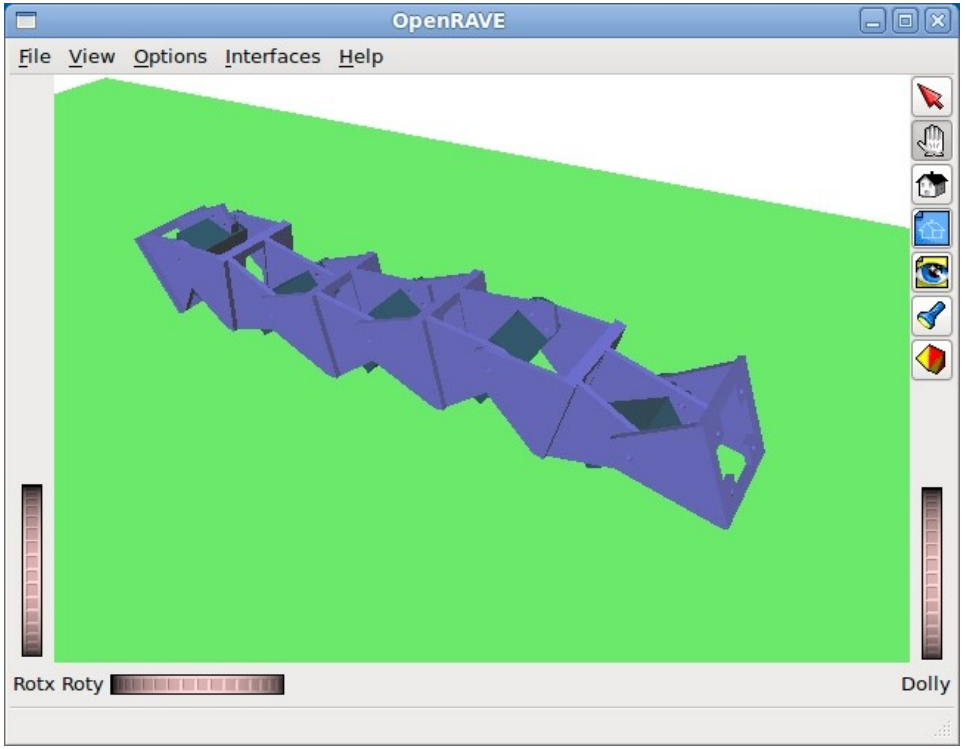
Cube Simulator



http://bit.ly/bnN4KP

Simulación (II)

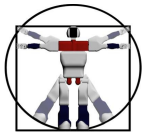
Demo



- Simulador: **OpenRave** + **OpenMR** plugin
- **OpenMR** = OpenRave Modular Robot plugin
- Más información: http://bit.ly/9a3fXk

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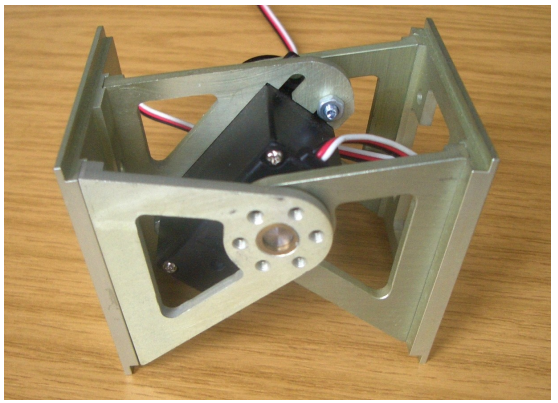
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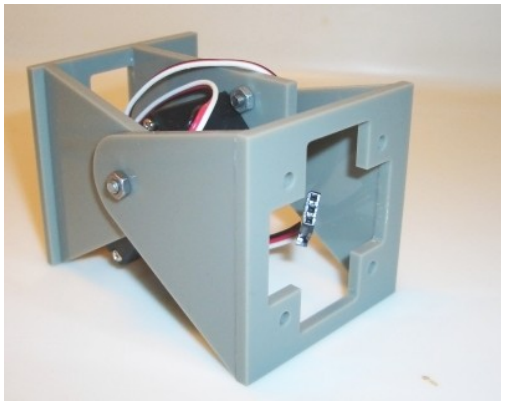
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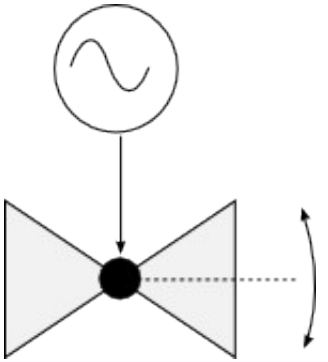
Conclusiones

El modelo basado en **generadores sinusoidales es válido** para la locomoción de robots modulares con topología de 1D

- Requiere muy pocos recursos para su implementación
- Se consiguen movimientos muy suaves y naturales
- Se pueden realizar diferentes tipos de movimientos
- Configuraciones mínimas de 2 y 3 módulos







$$\varphi_i(t) = A_i \sin\left(\frac{2\pi}{T} + \psi_i\right) + O_i$$

Trabajo futuro (I)

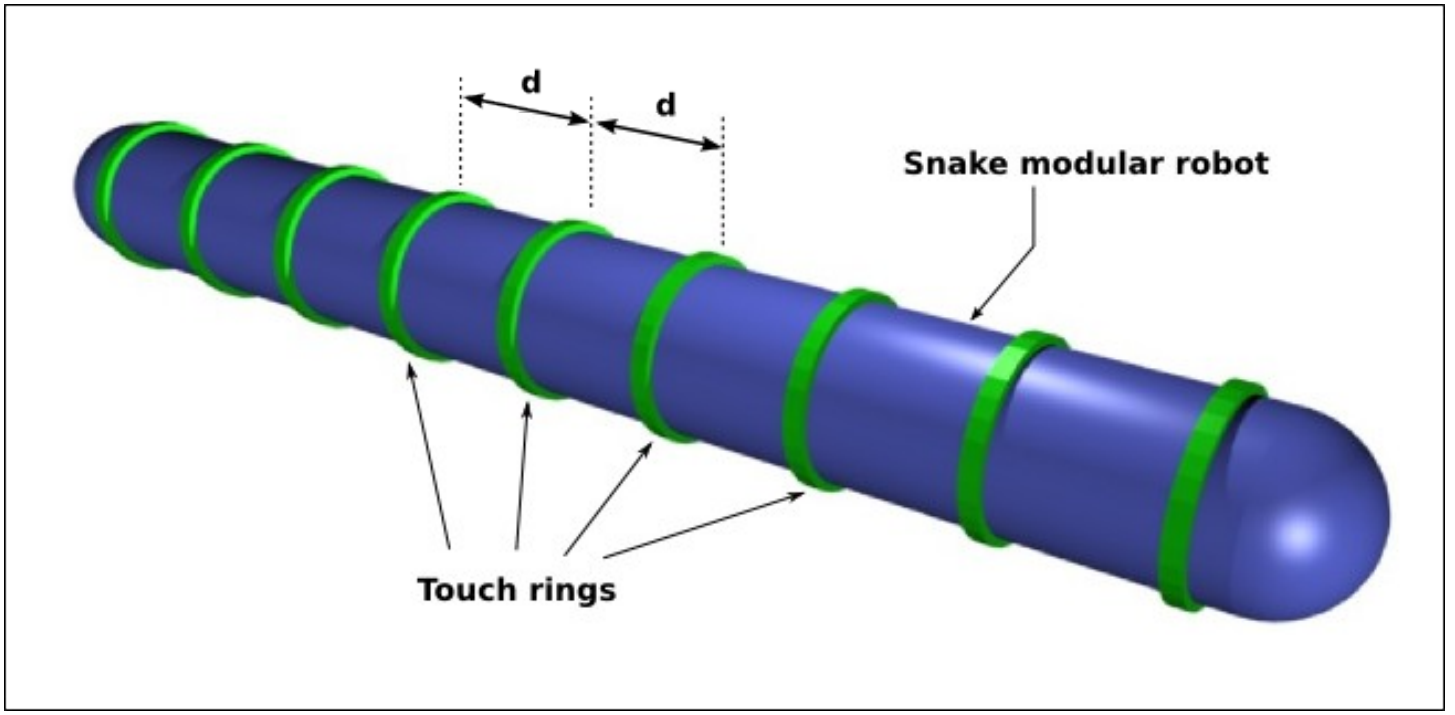




- **Aplicación**: Robots de búsqueda y rescate en zonas catastróficas

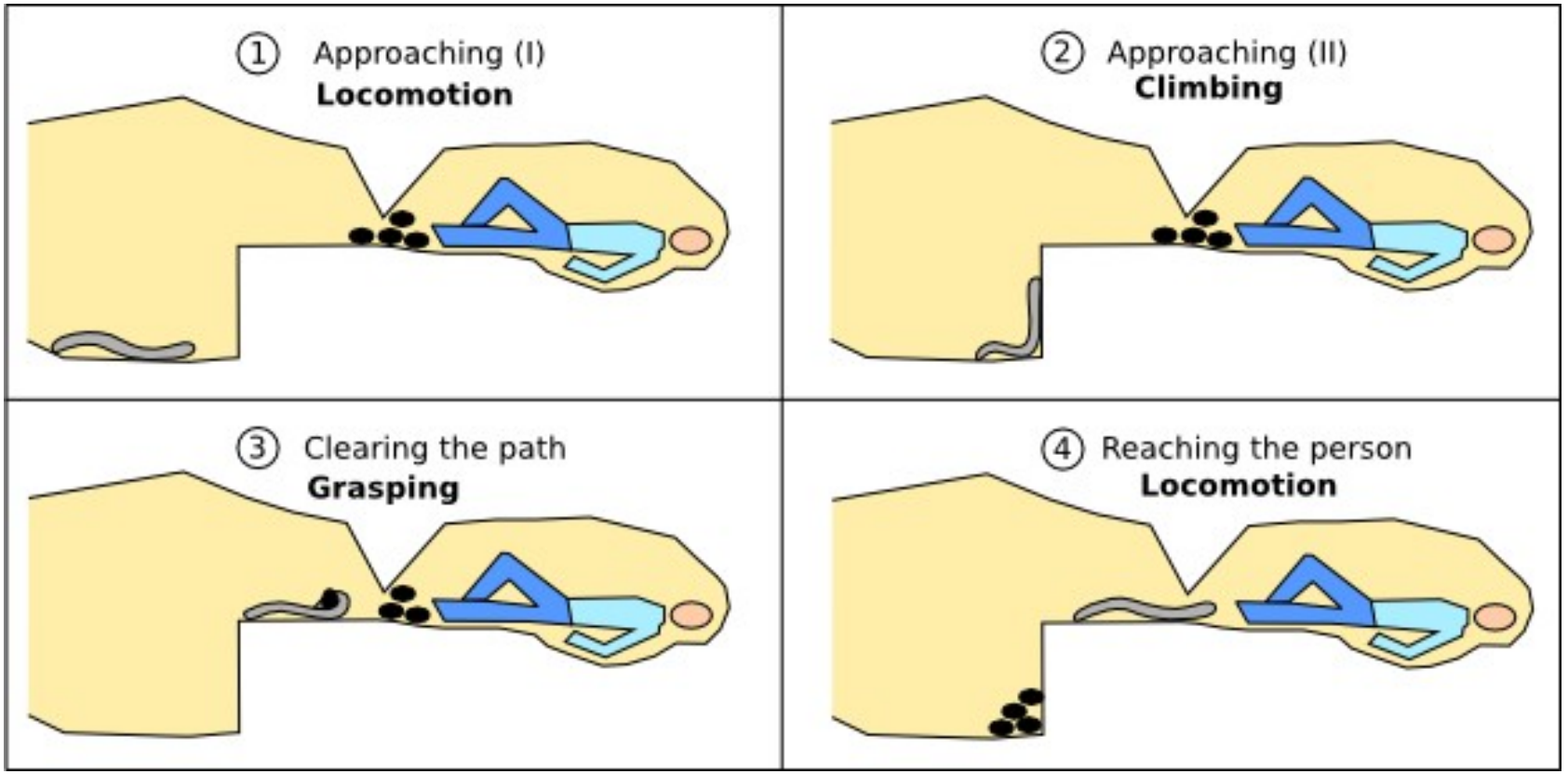
Juan Gonzalez-Gomez, Javier Gonzalez-quijano, Houxiang Zhang, Mohamed Abderrahim, "**Toward the sense of touch in snake modular robots for search and rescue operations**". In Proc of the ICRA 2010 workshop on modular robots: State of the art. May-3rd, Anchorage, Alaska

Trabajo futuro (II)



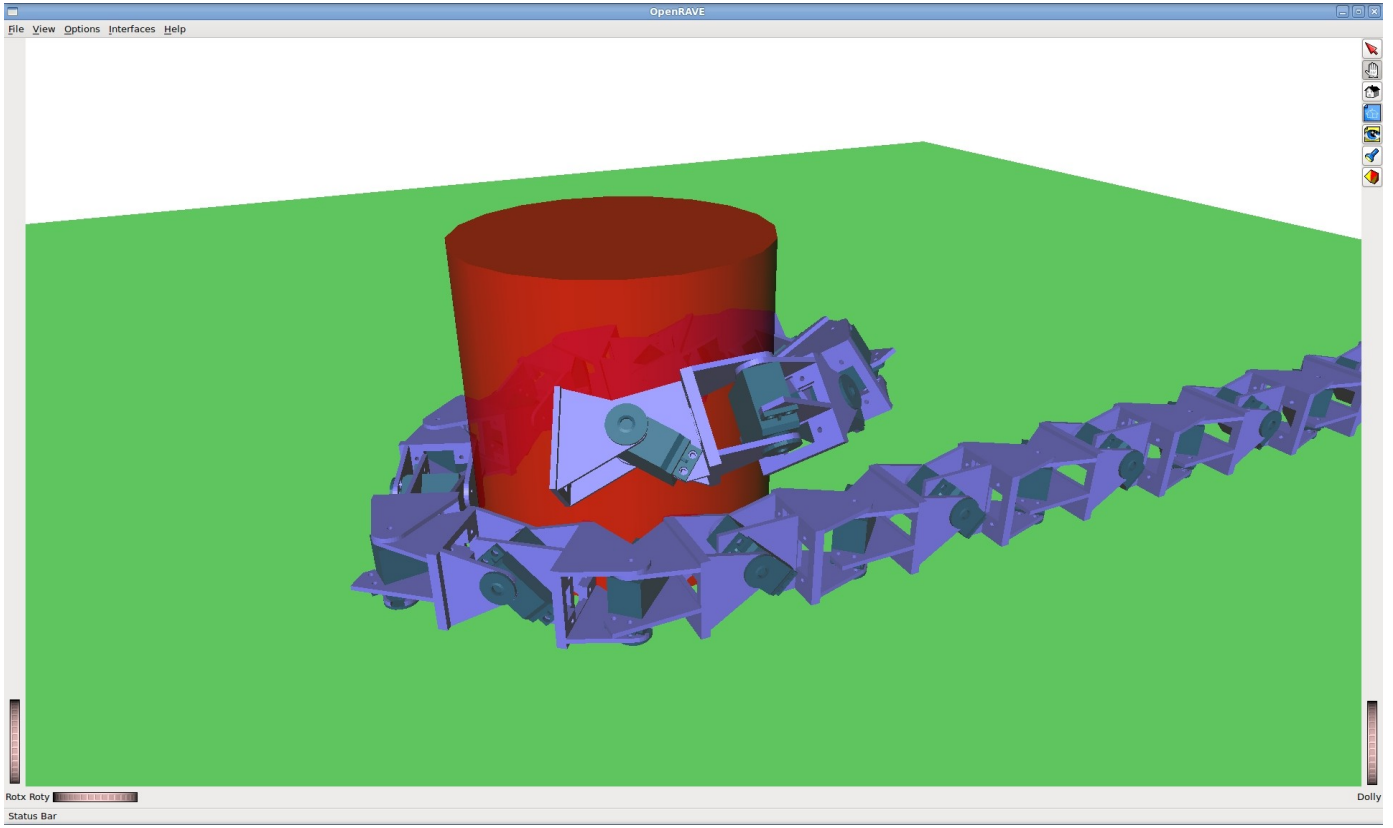
- Dotar del “sentido del tacto” a las serpientes modulares

Trabajo futuro (III)



- **Capacidades**: Locomoción, trepar y agarre de objetos

Trabajo futuro (IV)



- Agarre y manipulación de objetos con serpientes modulares

Trabajo futuro (V)





- Locomoción de otras configuraciones

Locomoción de Robots ápodos modulares



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My Master Thesis



- Title: Distributed Control Algorithm For A Multi Cellular Robotic Organism.
- Author: **Avinash Ranganath**
- Under the EU sponsored modular robotics project called SYMBRION.
- Objectives:
  - Develop a framework to control locomotion and obstacle avoidance behavior in modular robots.
  - Distributed controller.

ubuntu

My Master Thesis

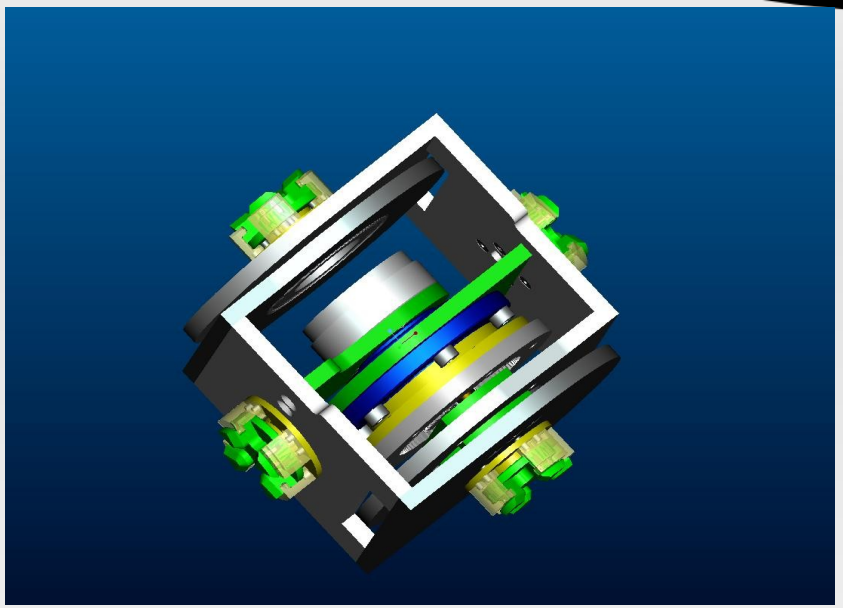


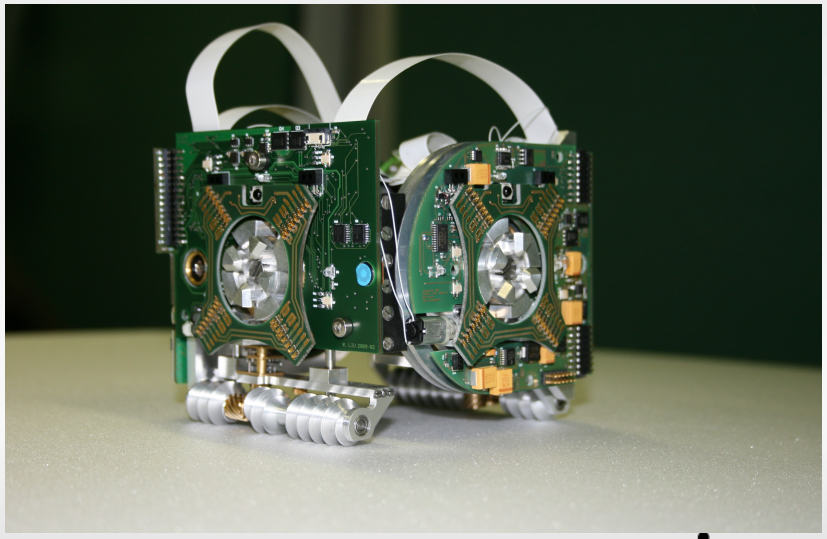
- Unified controller for different modular robotic configurations.
- Fault tolerance capability.

- My work is based on Digital Hormone Method [DHM], as proposed by *Shen et al.* from ISI, USC.

SYMBRION Module

- Homogeneous, open sided cubes.
- 2 interlocking 3D U shaped body.
- 1 motor with 1 DOF.
- 4 Connectors.
- Screw driver wheels.
- 1 tilt and multiple IR sensors.

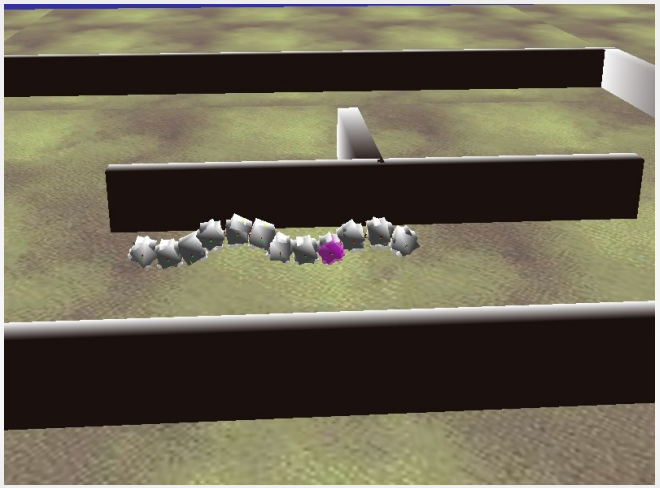


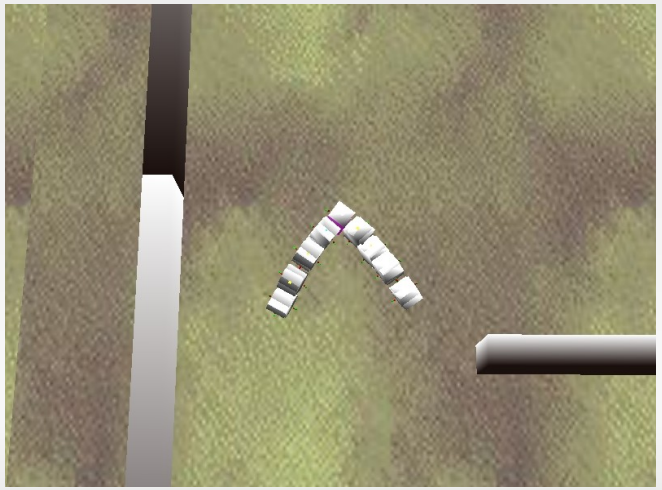


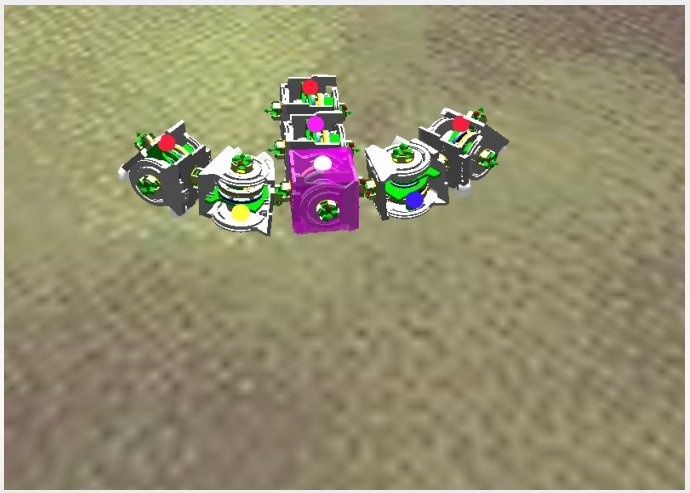
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Implementation Platform

- The framework was tested in a distributed simulation environment called Symbricator3D.
- I implemented the control algorithm on three different robotic organisms.







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Locomotion



- Coordinated local action of individual modules produces locomotion as a global behavior.
- Eg: Caterpillar locomotion gait is a sin wave.
  - Modules oscillate between +45 and -45 degrees.
  - Interval between the oscillations determine the wave length.
- So how do you get individual modules to perform local actions to produce the global behavior based on the organism they are a part of?

Inspiration of DHM



In multicellular biological organisms, there are various types of cells. Some of them generate and diffuse hormones, which are targeted are certain other types of cells. All cell types receive these hormone, but are reacted upon only by the designated type of cells.

-Wei Min Shen

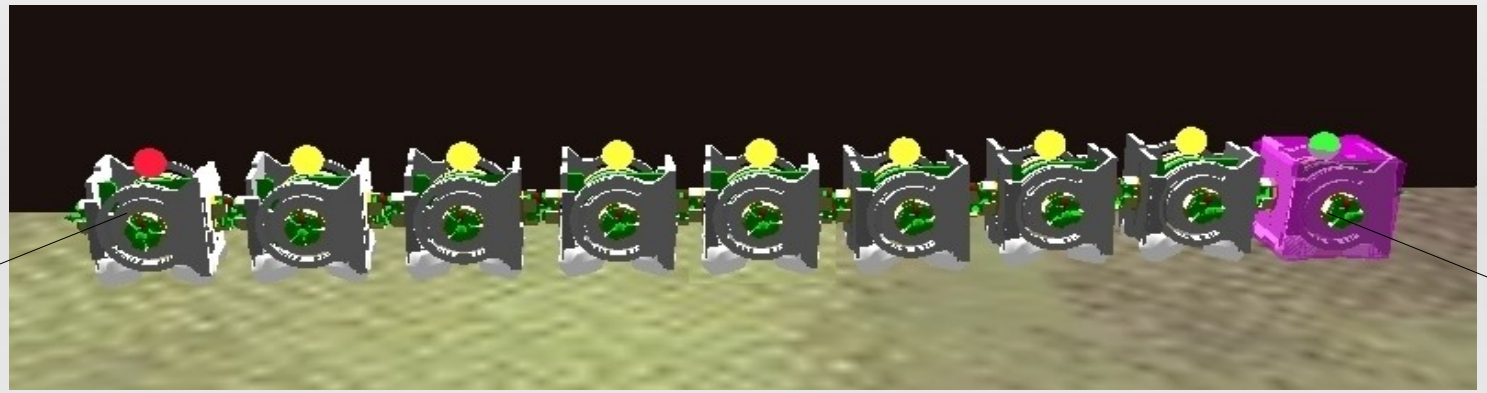
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Digital Hormone Method



- Topology Mapping - Where in the topology am I located?
- Local Communication - What are my neighbors doing?
- Environment Input - What does my sensor read about the local environment?
- Internal Variables - What are the values of my internal variables? Eg: Tilt sensor, Direction variable, etc.

Caterpillar Gait Using DHM



- Node_1: Rotates motor to +/- 45 degrees. Generate and initiate hormone diffusion.
- Node_2 to Node_n-1: Perform the same action as the parent node. Diffuse hormone.
- Node_n: Perform same action as its parent. No hormone diffusion.
- Node_1: After ´x´ amount of time, rotate motor in the opposite direction. Generate and initiate hormone diffusion.

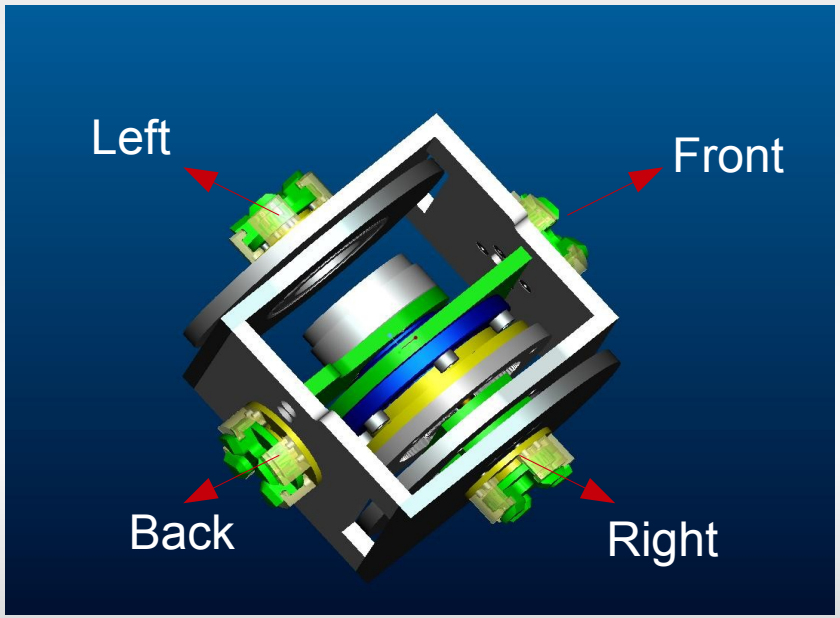
Caterpillar Gait Using DHM



So how does each node know whether or not it is responsible for initiating the hormone diffusion?

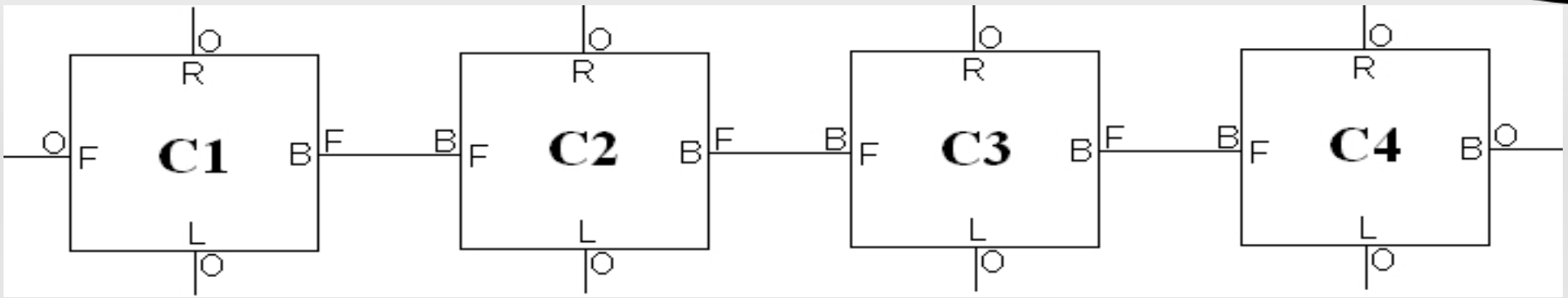
Topology Mapping – Module Type

- Three distinct module types in caterpillar configuration.
  - Tail: Node_1
  - Spine: Node_2 to Node_n-1
  - Head: Node_n
- Each module has four connectors.



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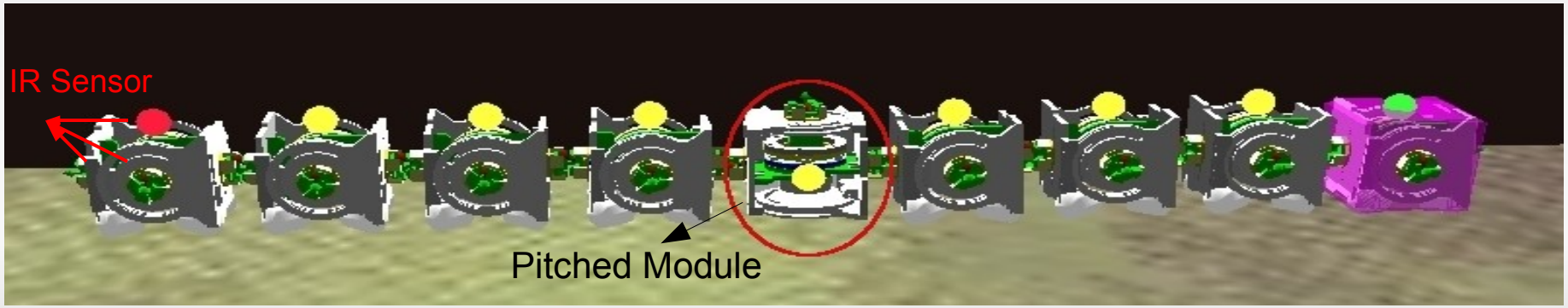
Topology Mapping – Module Type



- Connector: Can be connected to another connector in five different ways.
- Module: Can be connected to other modules around it in 5^4 = 625 different ways.
- Modules communicate connector information with neighbors.
- Calculate Level_0 topology mapping.
- Modules choose local action based on module type.

Obstacle Avoidance in Caterpillar

- Use IR sensors on the head node.
- Use pitched module for rotation.
- Moves back if pitch module not present.
- Head module generates ´Obstacle Hormone´.
- Reacted upon either by the pitched or the tail module.

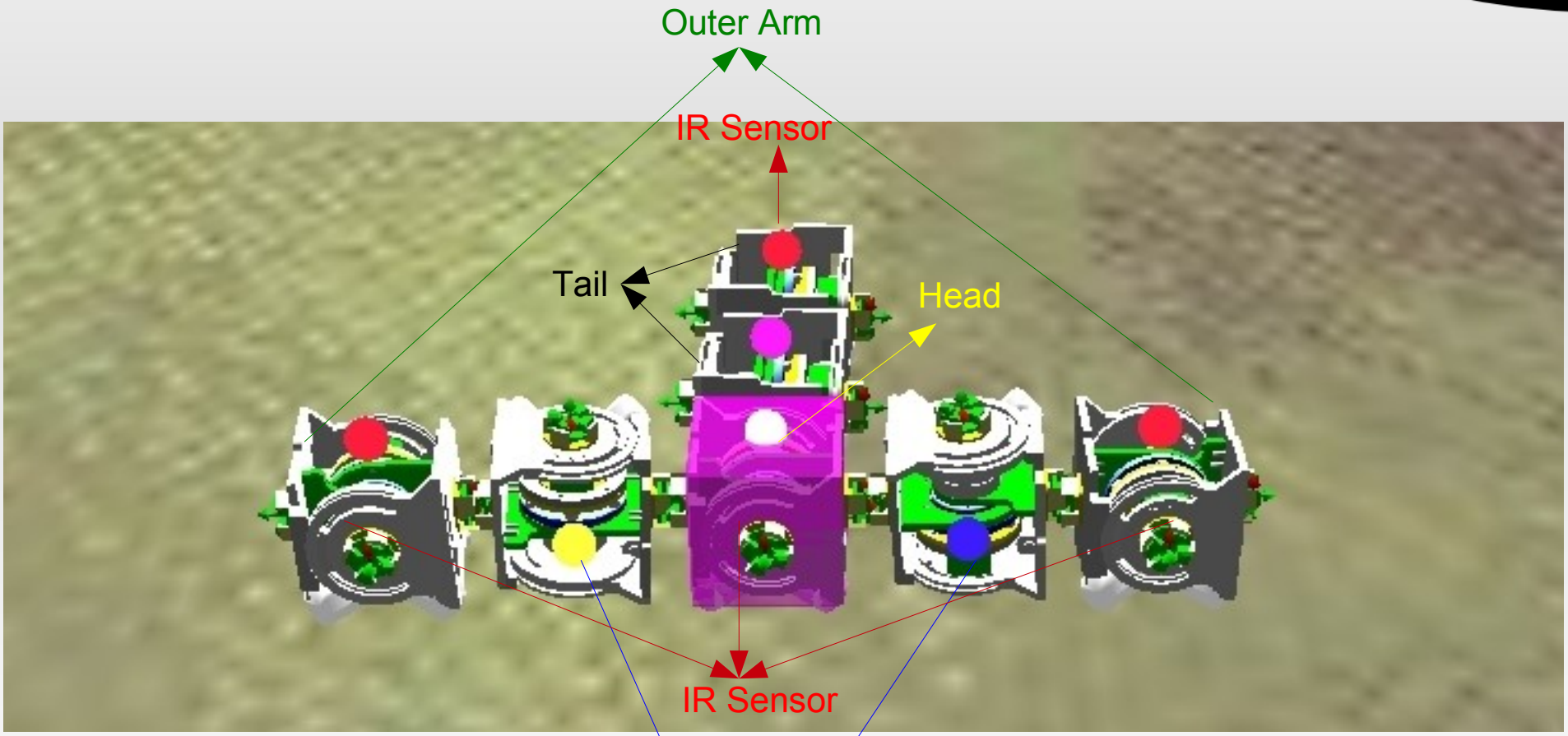


Obstacle Avoidance in Caterpillar



- What if there is no pitched module?
  - Tail node becomes head node.
  - Head node becomes tail node.
  - Organism moves in the reverse direction.

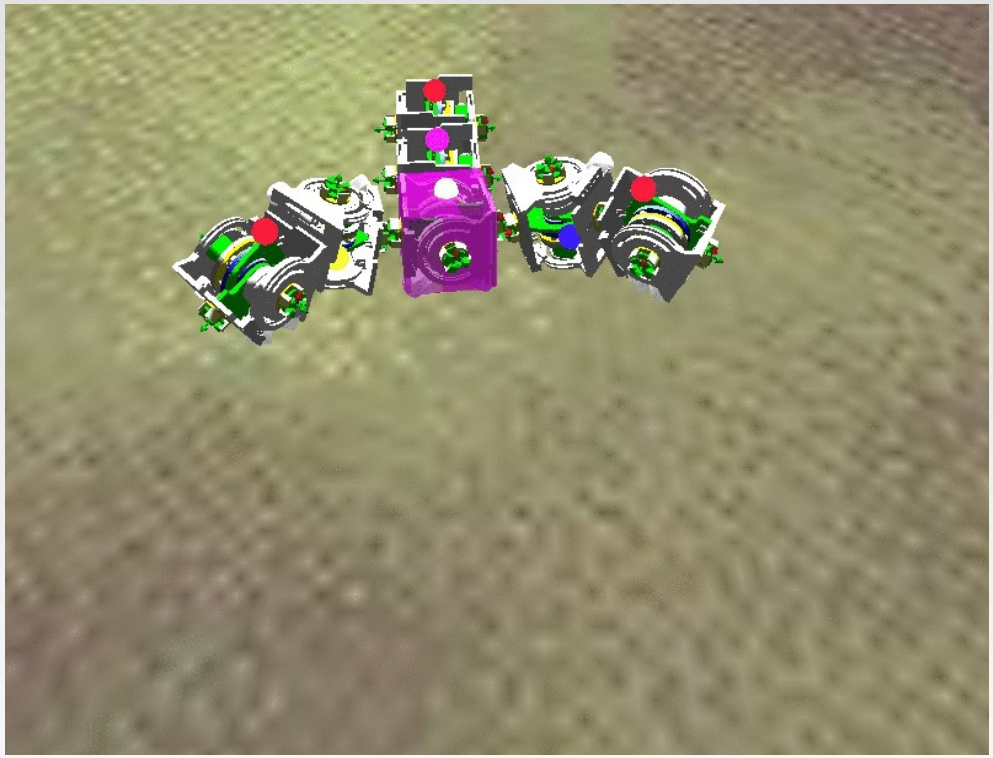
Scorpion Organism



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Scorpion Locomotion Gait

- Forward Motion
  - Lift arm up.
  - Swing arm forward.
  - Push arm down.
  - Swing arm backward.



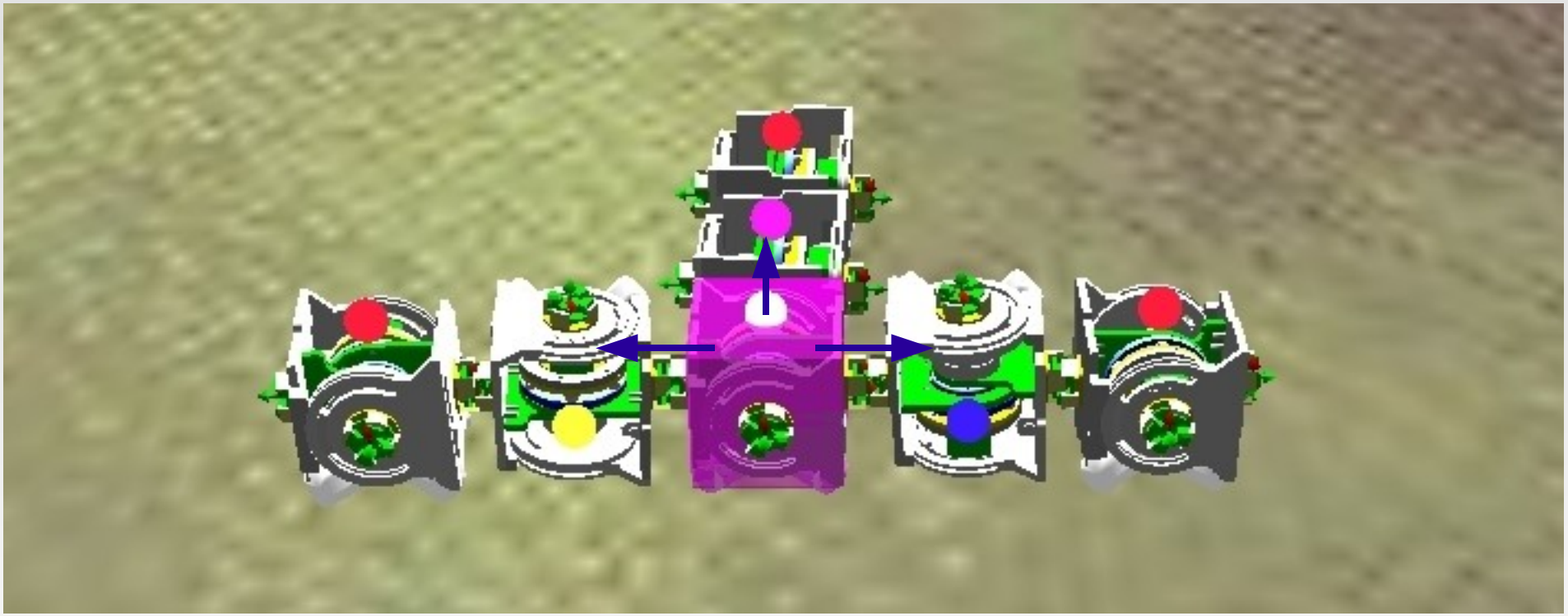
Outer arm module
Inner arm module

Scorpion Locomotion Gait

- Backward Motion
  - Lift arm up.
  - Swing arm backward.
  - Push arm down.
  - Swing arm Forward.
- Turn Right
  - Left arm forward.
  - Right arm backward.
- Turn Left
  - Right arm forward.
  - Left arm backward.

Outer arm module
Inner arm module

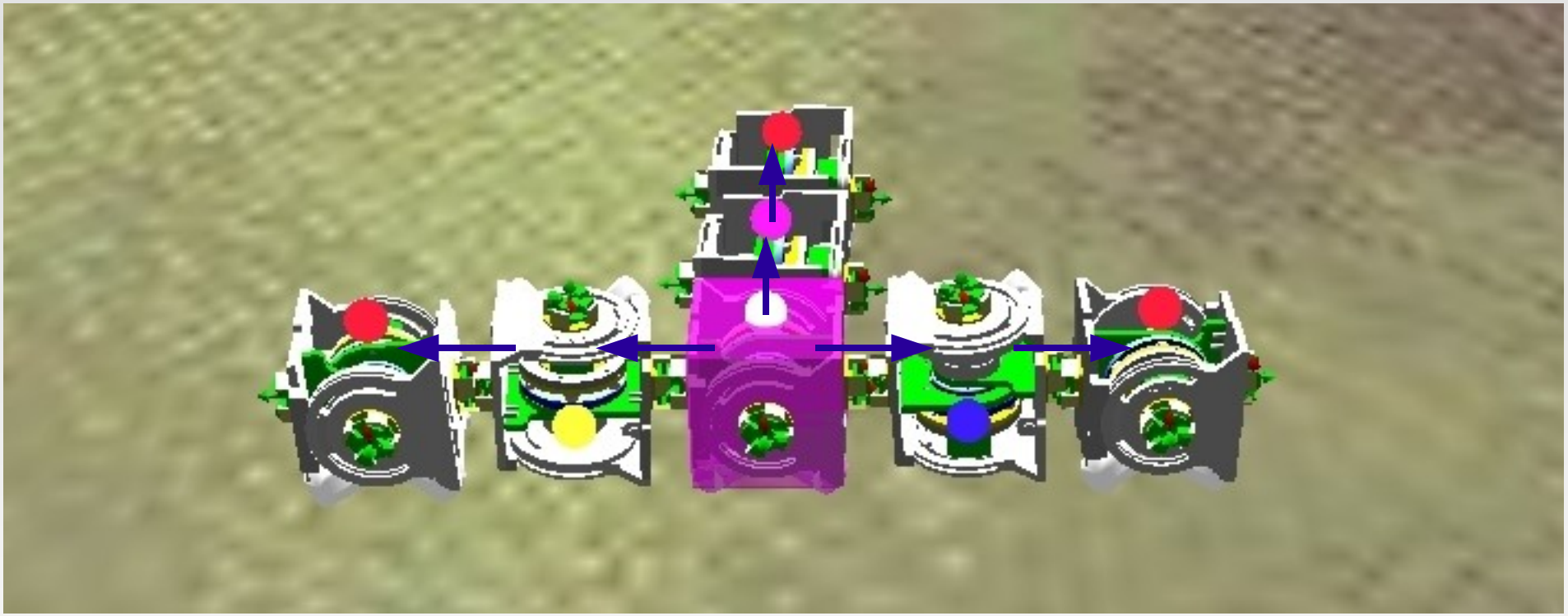
Scorpion Gait - DHM



Hormone generated by Head node

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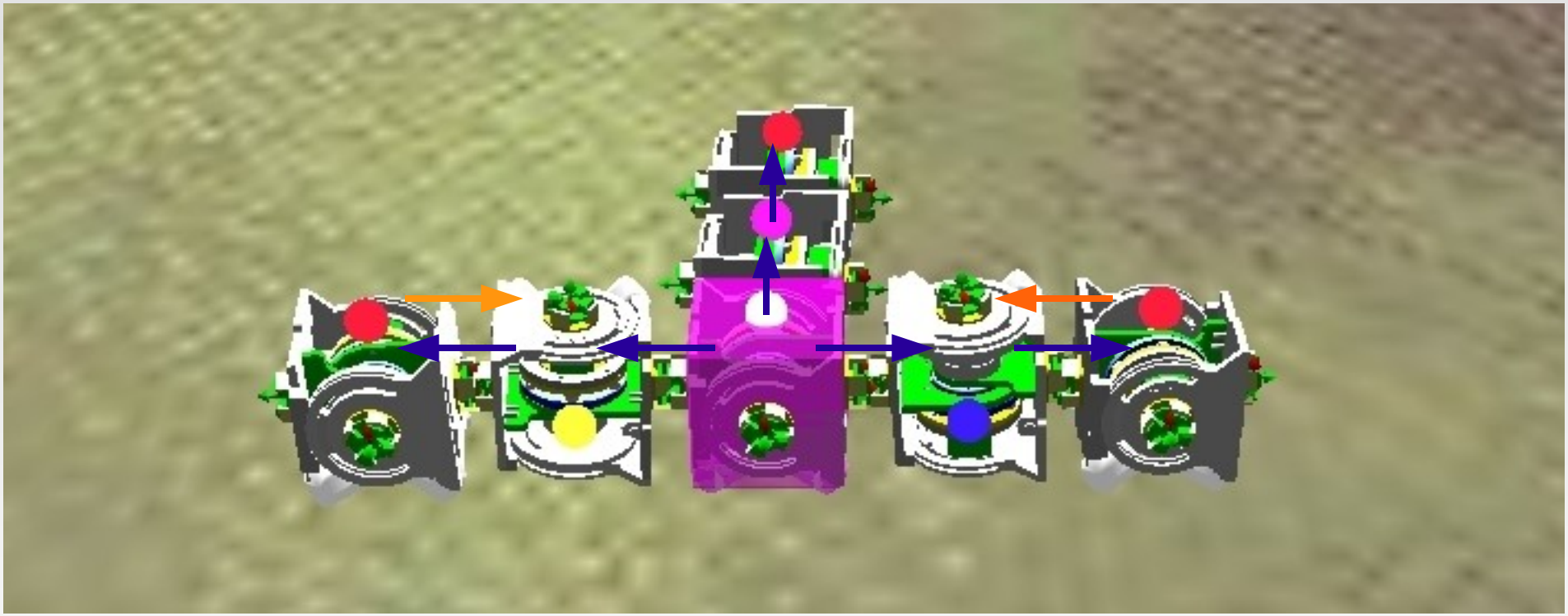
Scorpion Gait - DHM



Hormone generated by Head node

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Scorpion Gait - DHM

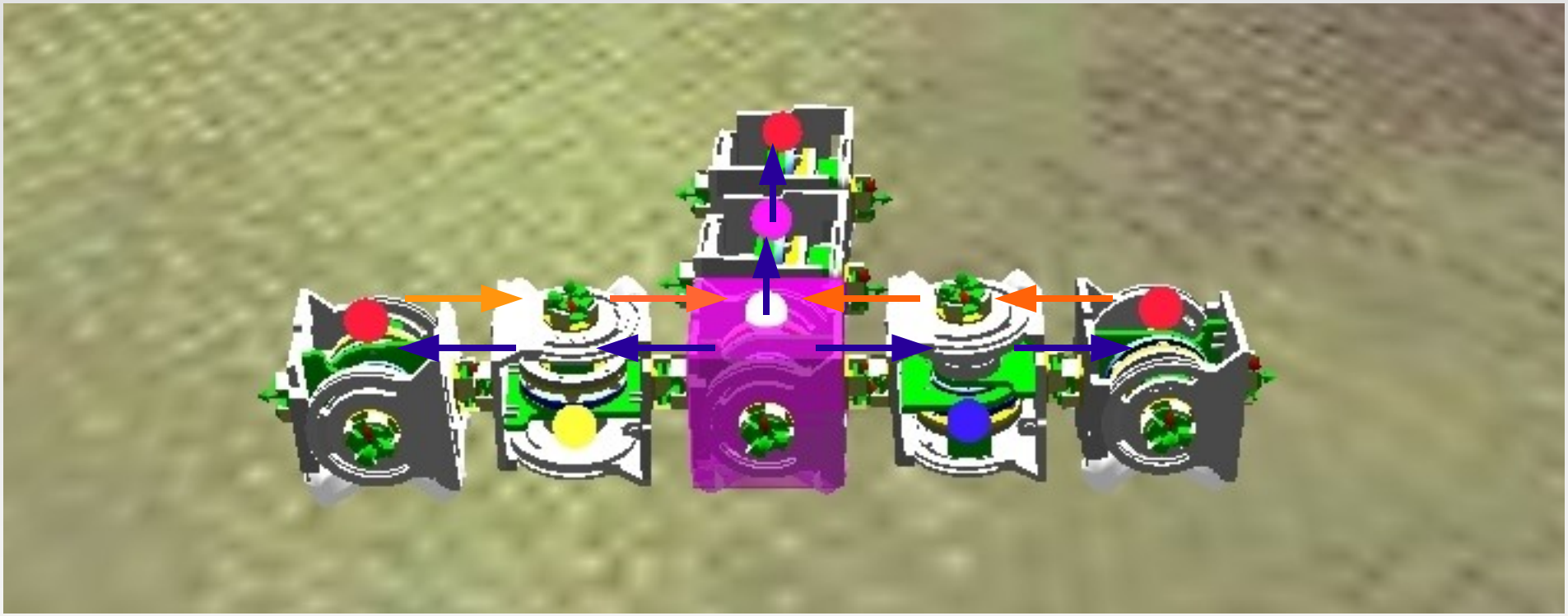


Hormone generated by Head node

Hormone generated by Outer Arm node

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Scorpion Gait - DHM

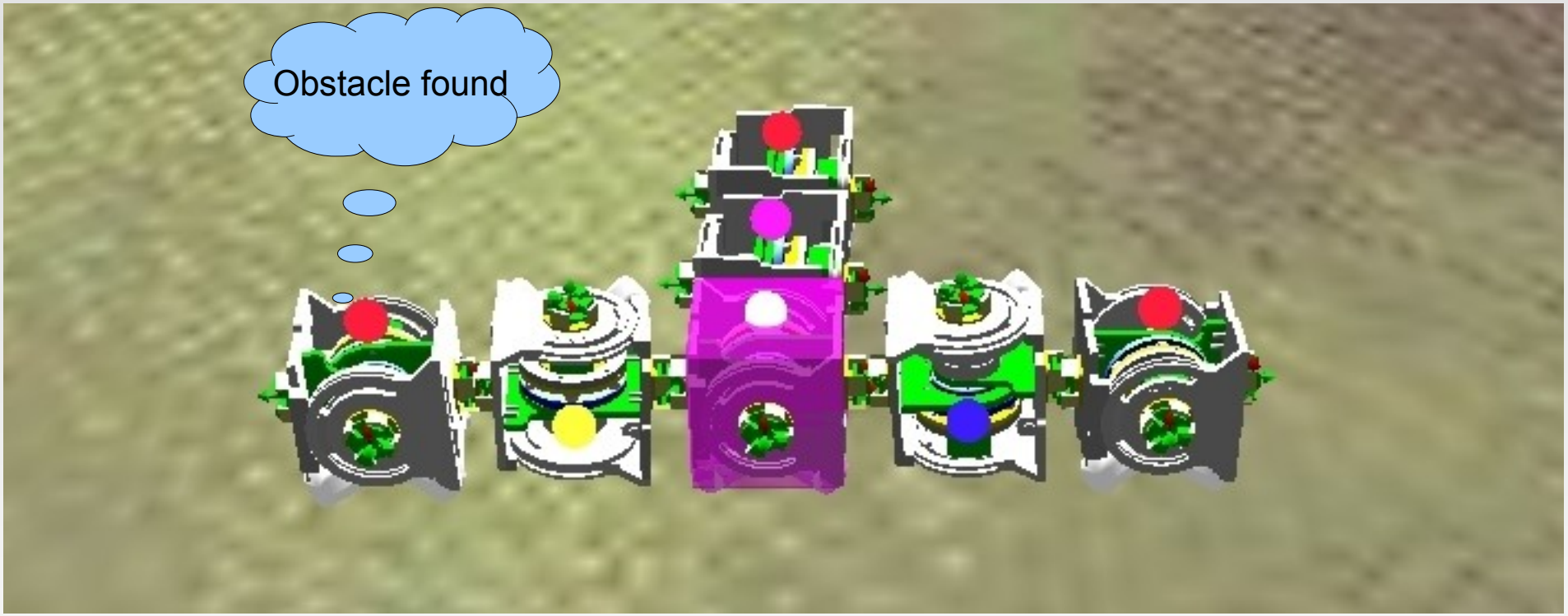


Hormone generated by Head node

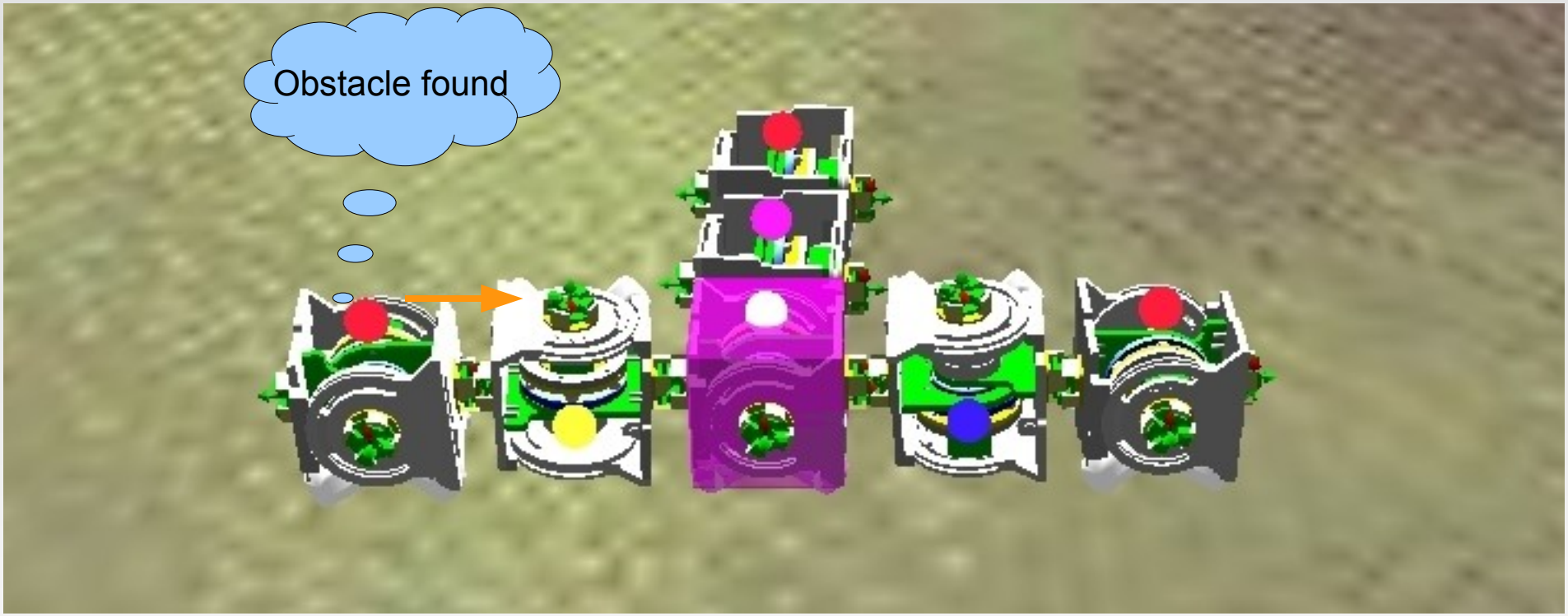
Hormone generated by Outer Arm node

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Scorpion Obstacle Avoidance - DHM



Scorpion Obstacle Avoidance - DHM



→ Hormone generated by Outer Arm node

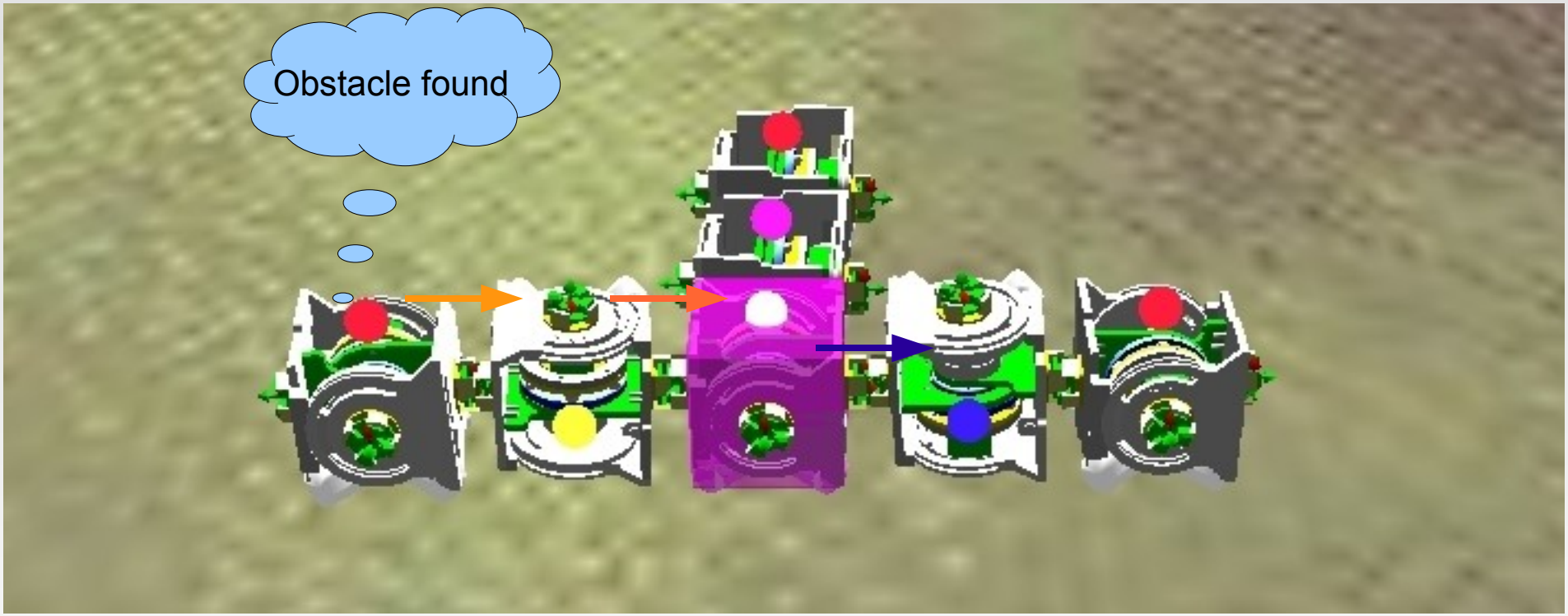
Scorpion Obstacle Avoidance - DHM



→ Hormone generated by Outer Arm node

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Scorpion Obstacle Avoidance - DHM

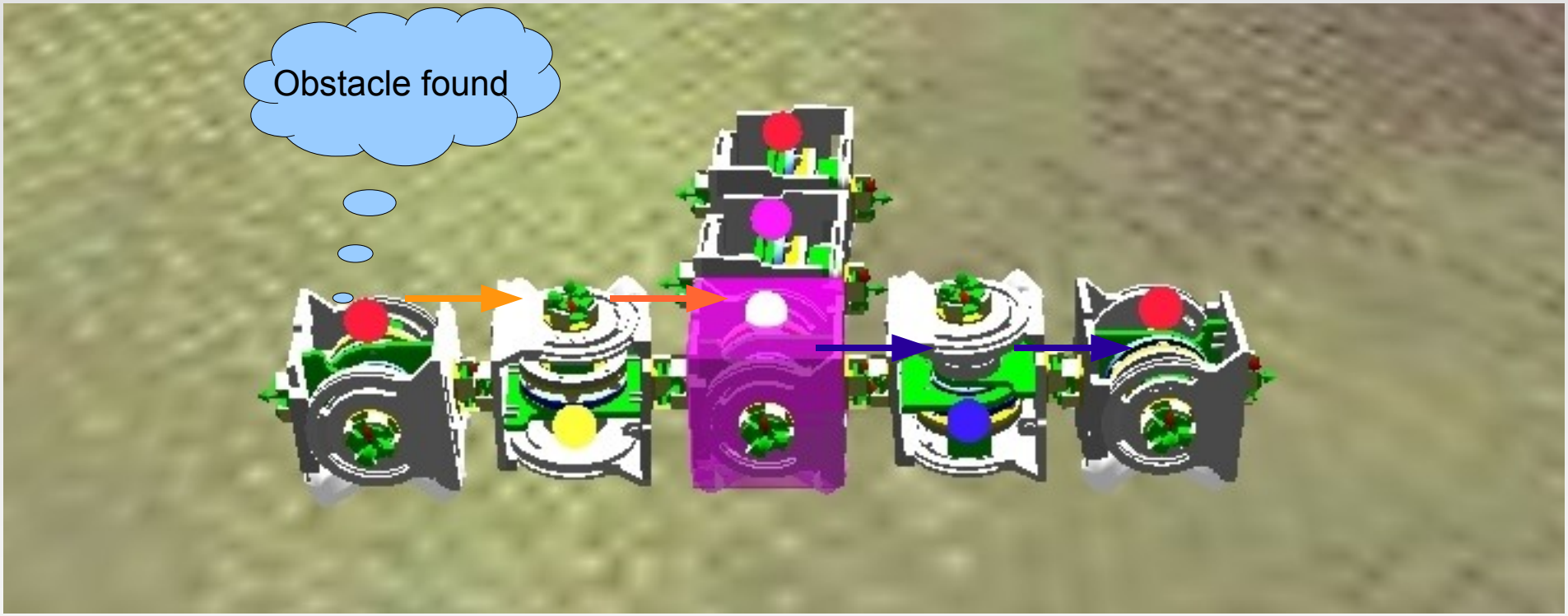


Hormone generated by Head node

Hormone generated by Outer Arm node

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Scorpion Obstacle Avoidance - DHM

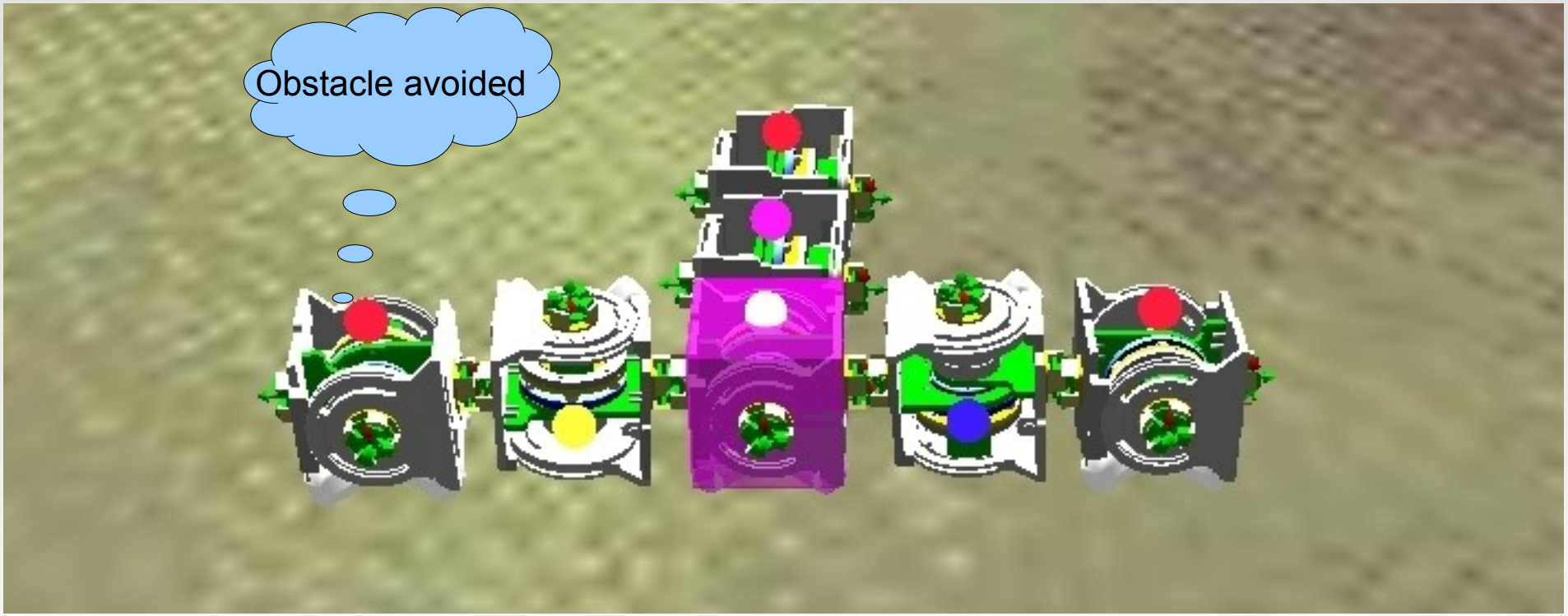


Hormone generated by Head node

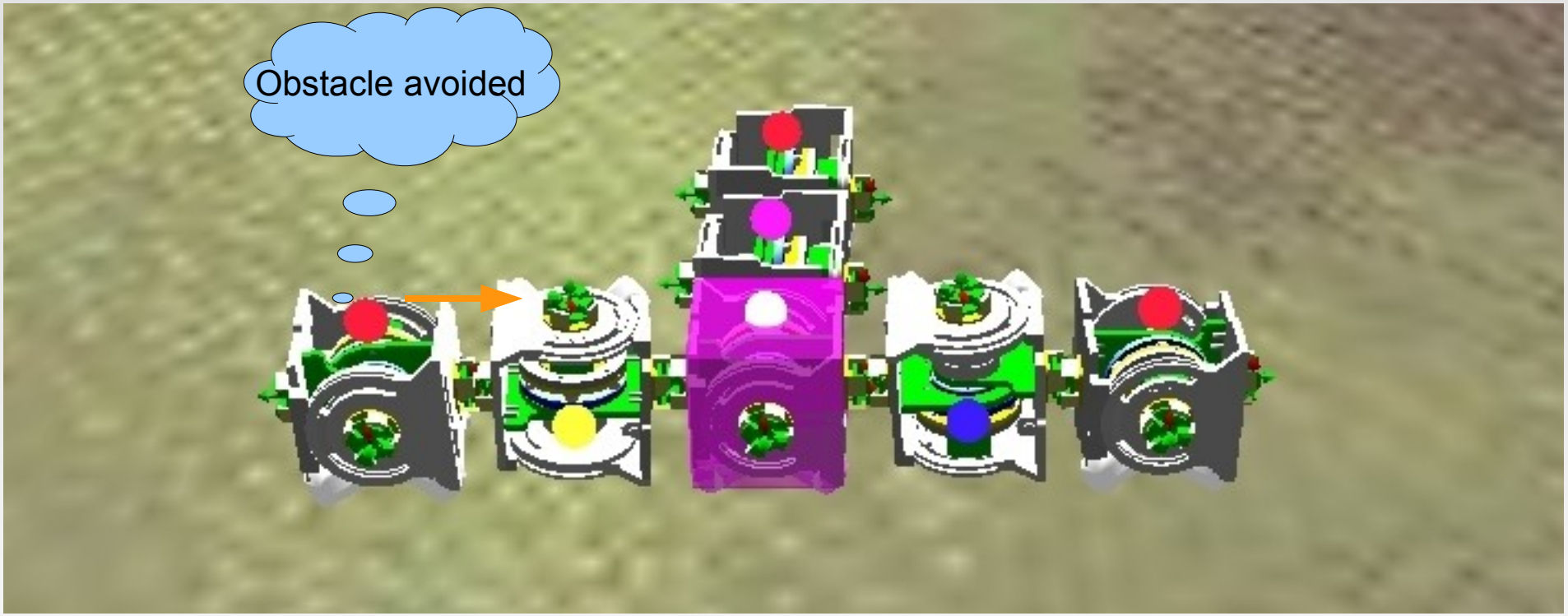
Hormone generated by Outer Arm node

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Scorpion Obstacle Avoidance - DHM



Scorpion Obstacle Avoidance - DHM



Hormone generated by Outer Arm node

Scorpion Obstacle Avoidance - DHM



→ Hormone generated by Outer Arm node

Scorpion Obstacle Avoidance - DHM



Hormone generated by Head node

Hormone generated by Outer Arm node

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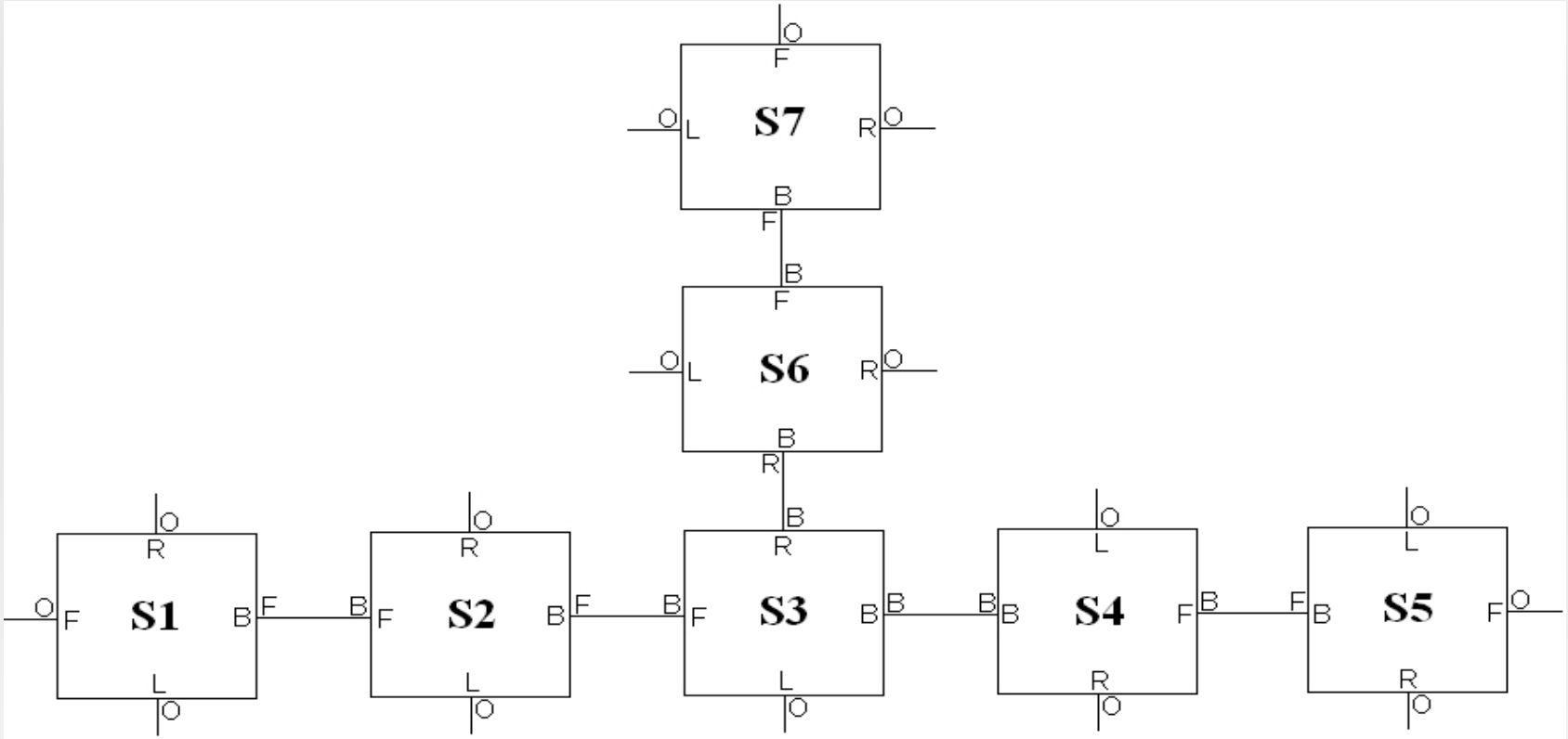
Scorpion Obstacle Avoidance - DHM



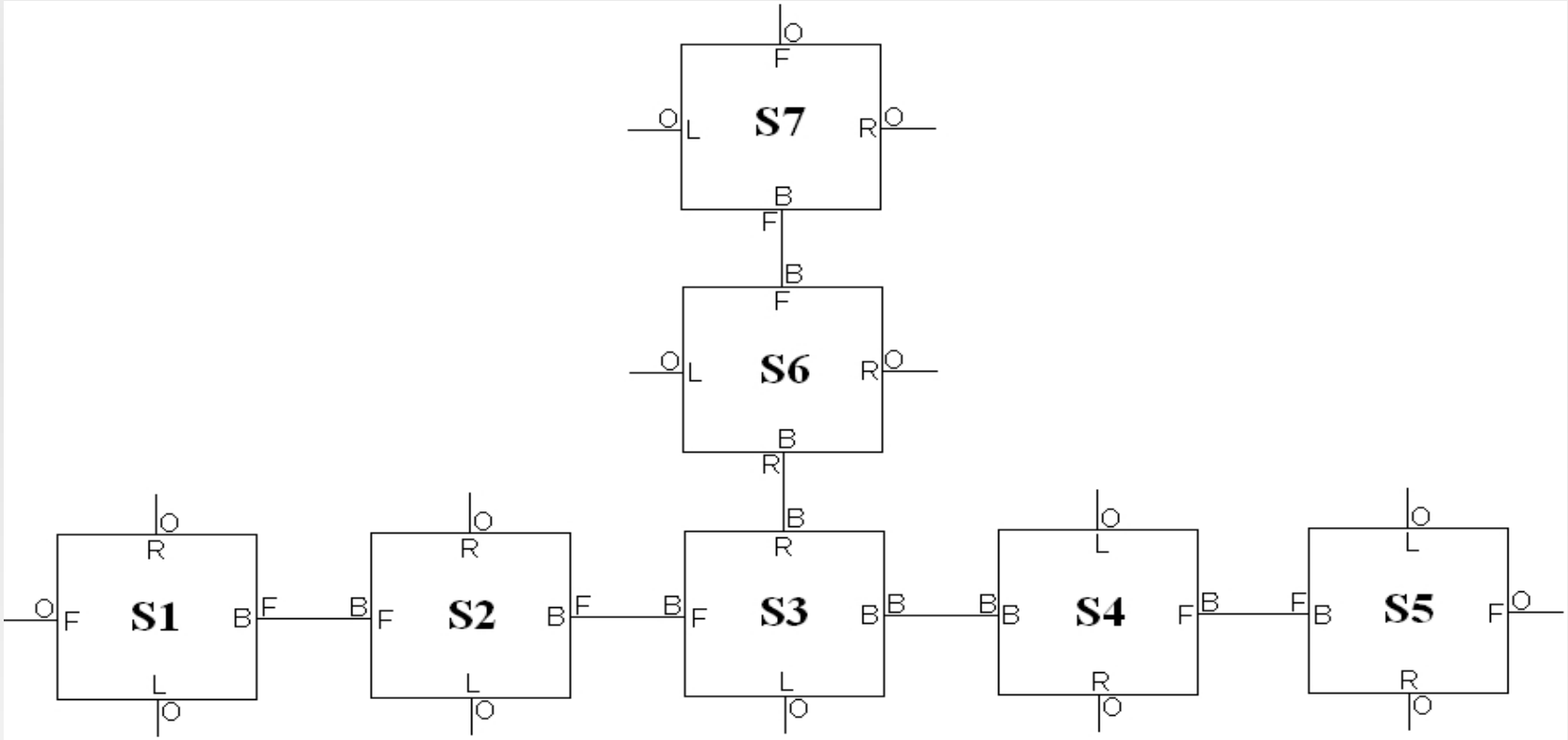
Hormone generated by Head node

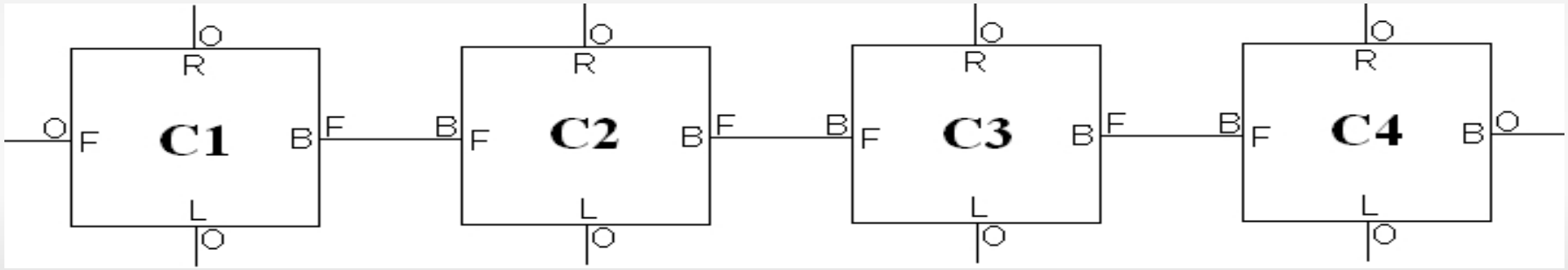
Hormone generated by Outer Arm node

Scorpion Topology



Scorpion Topology



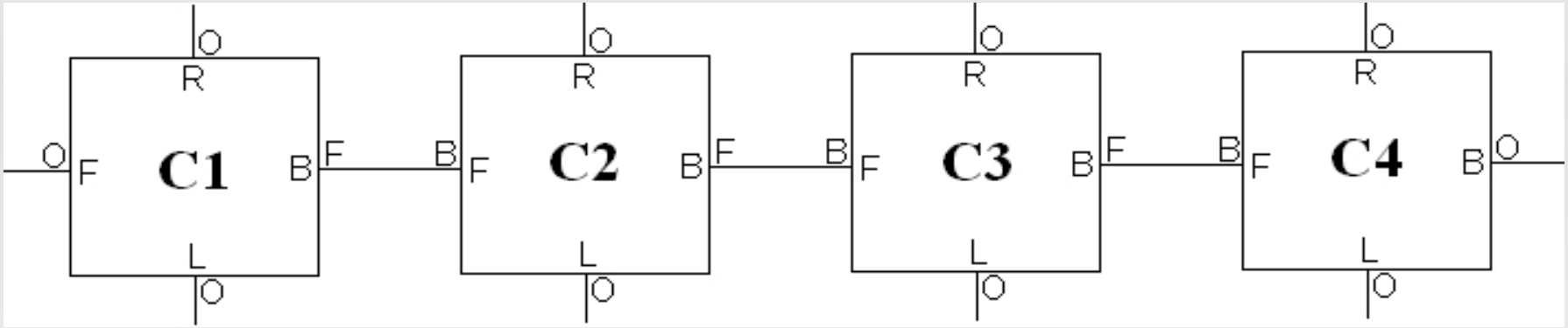


Multi Level Topology Mapping



- Level-0 Mapping
  - How a module is connected to each directly connected module.
- Level-1 Mapping
  - How a module is connected to modules that are at one module´s distance away.
- Level-n Mapping
  - Constrained to available memory.

Level-2 Topology Mapping

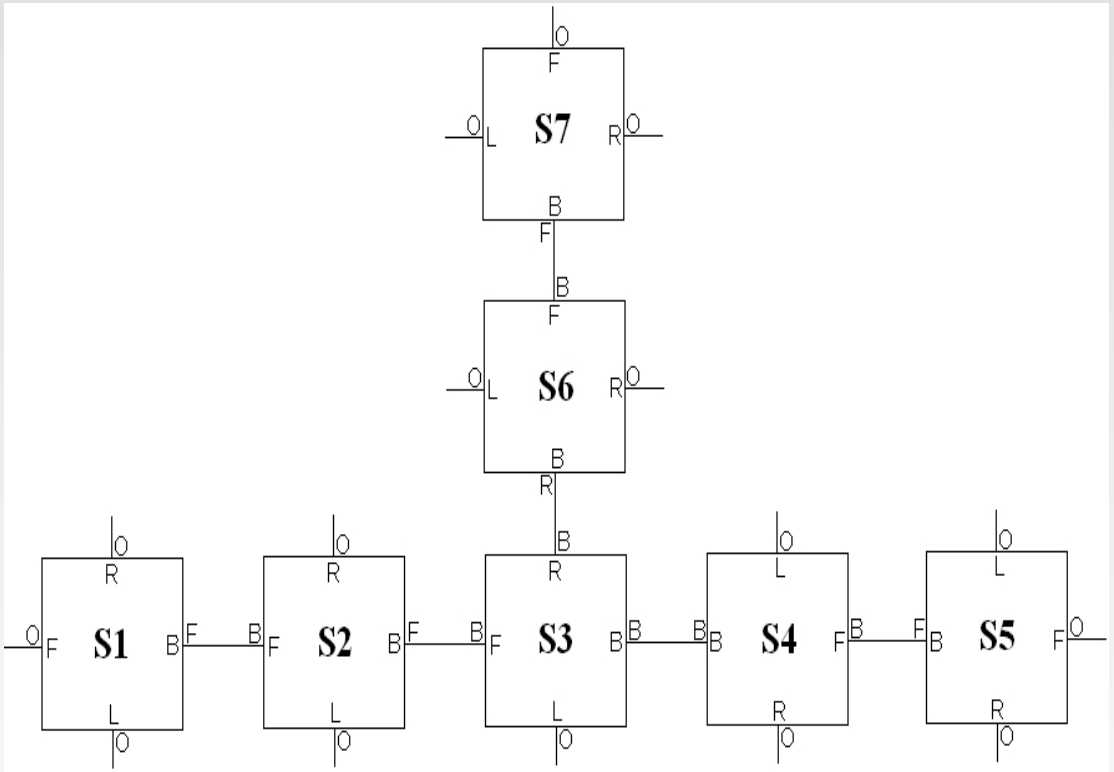


- C1
  - C1 to C4: {(B,F), (B,F), (B,F)}

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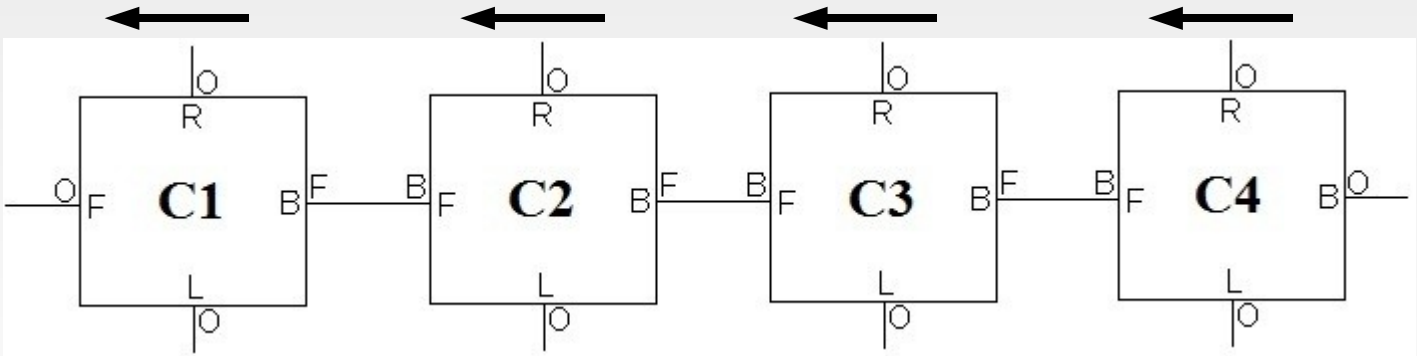
Level-2 Topology Mapping

- S1
  - S1 to S4: {(B,F), (B,F), (B,B)}
  - S1 to S6: {(B,F), (B,F), (R,B)}
- S5
  - S5 to S2: {(B,F), (B,B), (F,B)}
  - S5 to S6: {(B,F), (B,B), (R,B)}
- S7
  - S7 to S2: {(B,F), (B,R), (F,B)}
  - S7 to S4: {(B,F), (B,R), (B,B)}



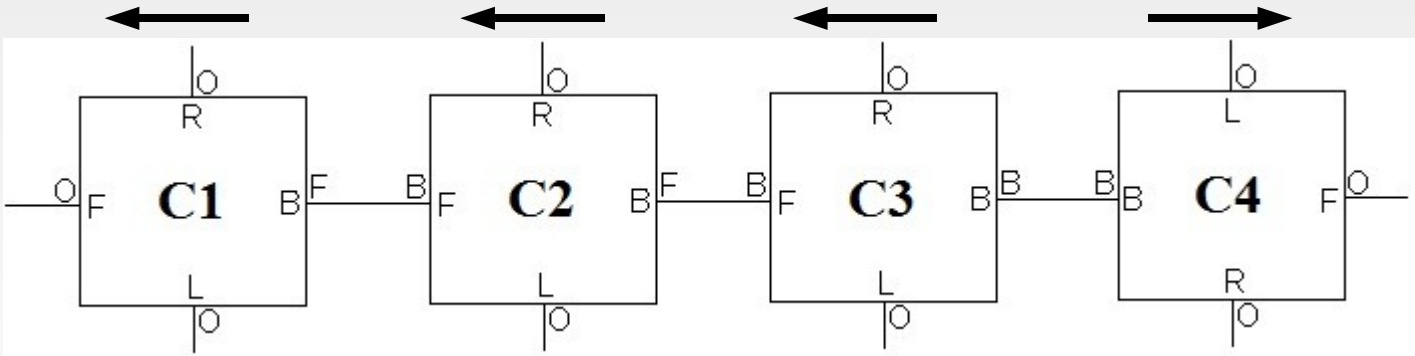
Drawbacks of the System

- Configuration specific.



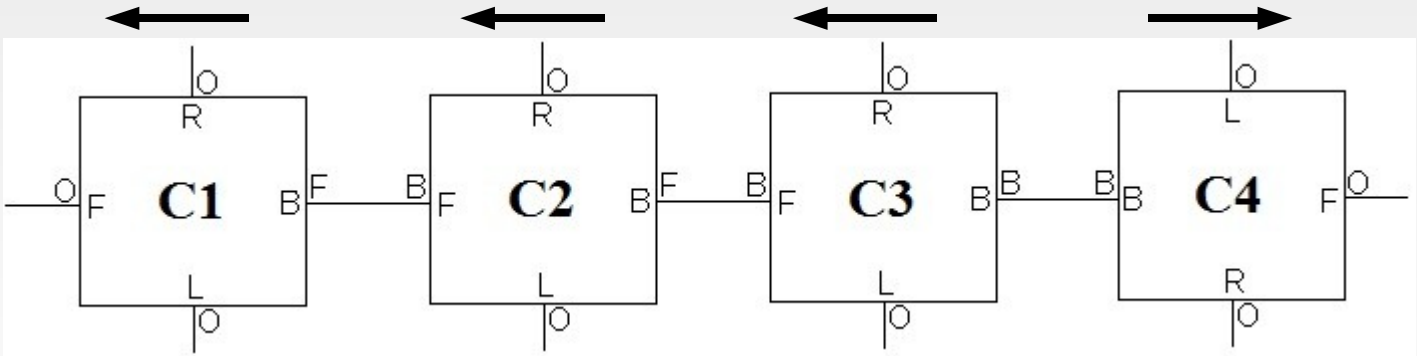
Drawbacks of the System

- Configuration specific.



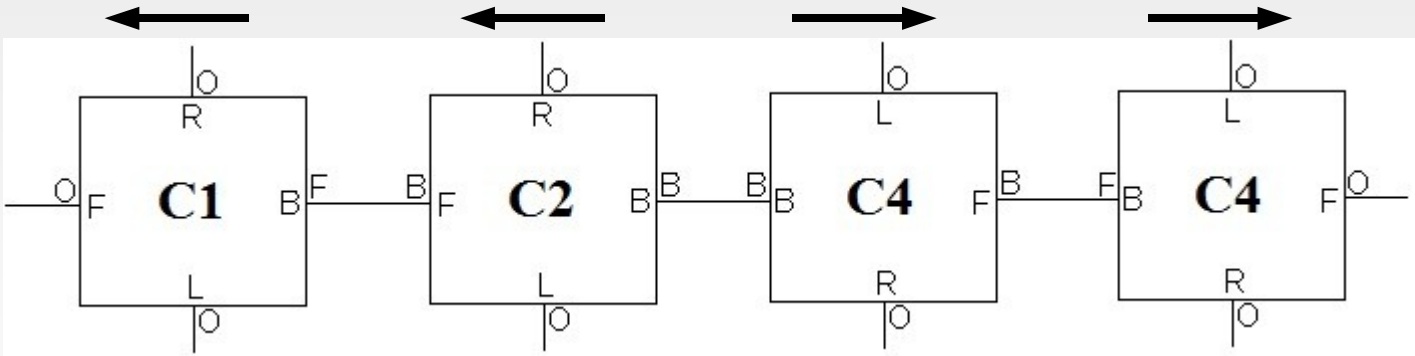
Drawbacks of the System

- Configuration specific.
- Does not work for all configurations.



Drawbacks of the System

- Configuration specific.
- Does not work for all configurations.



Drawbacks of the System

- Configuration specific.
- Does not work for all configurations.
- The underlying rules and parameter for the locomotion gait are hand coded.

What Next?



- Evolution of locomotion in higher order [3D] organisms.
- Investigate search and exploration techniques.
- Object (recognition) and manipulation.
- Use Juan´s robots by modifying it to include necessary sensors and communication channel.
- Look for a distributed simulation environment.

Questions



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